Mol-1: Open Frontier Biomolecular Intelligence

Molence Bio

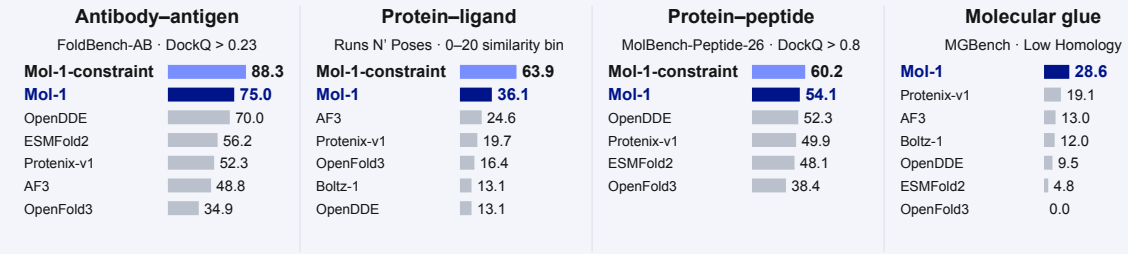
Abstract

We introduce **Mol-1**, a biomolecular foundation model for predicting molecular complexes and incorporating biological knowledge into those predictions. Mol-1 achieves frontier performance on structure prediction tasks spanning key modalities in drug discovery, including antibodies, small molecules, peptides, and molecular glues.

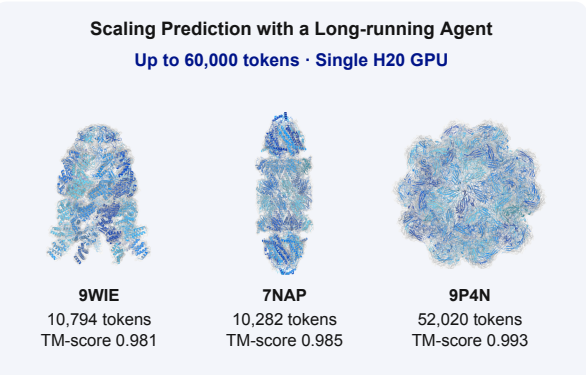
We put these predictive capabilities of Mol-1 to work in large molecular assembly and antibody design. Coupled with a task-specific workflow operated by a long-running agent, Mol-1 predicts assemblies of up to 60,000 tokens on a single NVIDIA H20 GPU, outperforming Anthropic's reported results on nine of ten evaluated assemblies. Using an earlier checkpoint of Mol-1, we developed a *de novo* VHH design workflow that yielded binding hits across 17 targets, including GPCRs, costimulatory receptors, and cytokines. Beyond binding, selected designs showed receptor-modulating activity and favorable biophysical properties. BCMA designs achieved SPR affinities as low as 43 pM, while experimental feedback increased the functional hit rate of 4-1BB designs from 7% to 85% in a second round. These studies show how molecular predictions can support both complex structural modeling and experimentally tested design.

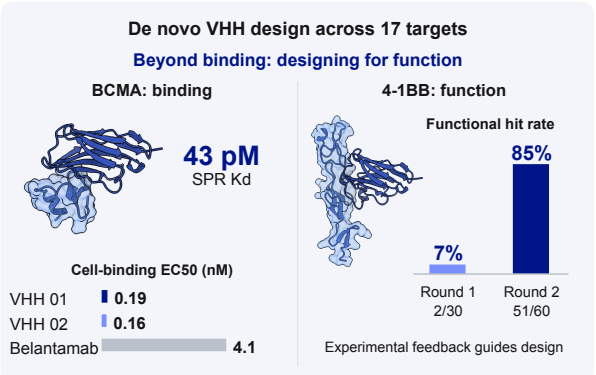
We release base Mol-1 on GitHub and host it on our web server, enabling researchers to evaluate the models independently, build on their capabilities, and pursue new biological questions.

01 Mol-1: A foundation to build on



02 How predictions are put to work





1. Introduction

Biomolecular structure prediction makes it possible to investigate molecular interactions at a scale that experiments alone cannot readily achieve (Abramson et al., 2024; ByteDance AML AI4Science Team et al., 2025; Chai Discovery Team, 2024; The OpenFold3 Team, 2026; Wohlwend et al., 2024). Their value lies both in the structures they predict and in the scientific questions they enable researchers to explore. We introduce Mol-1, a biomolecular foundation model for scientific inquiry: *a model that researchers can build on, guide with biological knowledge, and use to formulate experimentally testable hypotheses.*

A Foundation to Build On. Mol-1 achieves strong performance across antibody–antigen, protein–ligand, protein–peptide, and molecular-glue complexes. The development of Mol-1 combines increased representational capacity with training over larger molecular contexts and greater recycling depths. Beyond prediction accuracy, we examine how effectively the model uses additional inference computation. Recycling deepens the refinement of a prediction, while sampling explores alternative candidates.

Biological modeling rarely begins without prior information. An experiment may identify an interaction site; a study may suggest a particular conformational state (Ward et al., 2013). Mol-1-constraint offers a way to use such evidence, as demonstrated in earlier restraint-guided models (ByteDance AML AI4Science Team et al., 2025; Chai Discovery Team, 2024; Isomorphic Labs Team, 2026; Passaro et al., 2025). Benchmarks measure the gains from supplied interface annotations and their sensitivity to incomplete or incorrect evidence. Case studies make these effects concrete, showing how a few epitope residues can guide the model toward binding poses that it otherwise fails to recover.

Putting predictions to work. The reach of a biomolecular model depends not only on what it can predict, but on how its predictions are put to work. We explore two ways to build on this foundation: a long-running agent for large molecular assemblies, and a VHH design workflow developed using an earlier checkpoint of Mol-1. These results show how *a strong molecular predictor, coupled with a workflow that can build on and challenge its predictions, can address biological questions beyond a single model call.*

Large molecular assemblies offer a demanding test: local interactions must be recovered accurately and reconciled within a coherent global structure. We build on Mol-1's subcomplex predictions with an agent developed specifically for this task, combining confidence-guided subcomplex selection, progressive assembly, and structural evaluation. A long-running agent operates this workflow, using inconsistencies in candidate assemblies to direct further prediction and revision. Mol-1 supplies the structural hypotheses; the workflow provides the means to test how they fit together and determine where additional modeling is needed. Running on a single NVIDIA H20 GPU, the combined system predicts assemblies of up to 60,000 tokens and outperforms Anthropic's reported results (Claude Science et al., 2026) on nine of the ten assemblies evaluated.

Recent advances in *de novo* antibody design have made obtaining binders increasingly tractable (Chai Discovery Team, 2025a; Didi et al., 2026; Kong et al., 2025; Mille-Fragoso et al., 2025; Pacesa et al., 2025; Peng et al., 2026; Ren et al., 2025; Stark et al., 2025; Team, 2025; Team et al., 2025; Wu et al., 2026; Zambaldi et al., 2024; Zhang et al., 2025). Using a VHH design workflow built on an earlier checkpoint of Mol-1, we obtained binding hits across 17 targets. However, binding alone does not establish whether designs achieve their intended molecular functions.

We therefore evaluated our VHH designs against predefined biochemical and pharmacological objectives. *De novo* BCMA-targeting VHHs achieved SPR-measured affinities as tight as 43 pM with a design budget of only 30 candidates and exhibited strong binding to cell-surface BCMA. Beyond high-affinity binding, designed VHHs demonstrated receptor-modulating activities, including 4-1BB agonism and CXCR6 antagonism. To our knowledge, Mol-1 also enabled the first experimental demonstration of *de novo*-designed VHHs that cross-react with two non-orthologous protein antigens as intended. Finally, by closing the loop between computational design and experimental validation, we increased the functional hit rate of 4-1BB VHHs from 7% to 85% in a second design round, illustrating the power of lab-in-the-loop optimization to steer *de novo* antibody design toward desired biological functions.

Open for further discovery. Mol-1 comprises a base model with a single checkpoint shared across tasks and molecular modalities, together with a fine-tuned constraint-conditioned version. All structure prediction results reported in this work use these two checkpoints. We release the base version of Mol-1 through GitHub[1] and provide access to both versions through a webserver[2]. Researchers can use these predictors to incorporate different sources of knowledge, develop their own workflows, and test hypotheses beyond those examined here.

2. Mol-1: A Foundation for Biomolecular Structure Prediction

We evaluate Mol-1, a biomolecular foundation model, on its ability to predict complex structures across diverse molecular modalities, e.g., antibody–antigen, protein–ligand, protein–peptide, and molecular-glue complexes. Information about binding sites defined by just a few residues further improves predictions through Mol-1-constraint. Below, we present benchmark results and structural examples across these molecular modalities. Details of the model architecture, training, and evaluation protocols are provided in Appendix A.

2.1. Antibody–Antigen Complexes

We first evaluate Mol-1 on four antibody–antigen complex structure prediction benchmarks (FoldBench-AB (Xu et al., 2026), AF3-AB (Abramson et al., 2024), PXMeter-AB-22to25 (Ma et al., 2025) and MolBench-AB-26), measuring the top-ranked success rate under the standard 5 seeds × 5 samples setting. For Mol-1, we report median success rates across five-seed subsets drawn from at least 32 seeds to reduce seed-selection variability. Here, success is defined across three standard DockQ thresholds (0.23, 0.49, and 0.80), capturing performance ranging from basic interface recovery to high-accuracy prediction (Figure 1a–d). At DockQ > 0.23, Mol-1 outperforms the strongest open-source baseline on each benchmark. Specifically, at this threshold, Mol-1 achieves a 75.0% success rate on FoldBench-AB and 57.3% on MolBench-AB-26 (see Appendix A for detailed evaluation protocols). Finally, to assess sensitivity to seed selection at a fixed 5-seed budget on FoldBench-AB, we evaluate all possible 5-seed combinations drawn from a pool of 32 seeds (Figure 1f). Notably, 62.22% of these combinations match or exceed the reported 75.0% success rate, confirming the robustness across a majority of seed selections.

Deeper recycling can also improve sampling efficiency, reducing the number of seeds needed to achieve comparable prediction accuracy. On MolBench-AB-26, success rates at DockQ > 0.23 are approximately 58.7–58.8% under three configurations (Figure 17): 30 cycles with 5 seeds,

[1] https://github.com/molence/Mol-1
[2] https://lab.molence.com

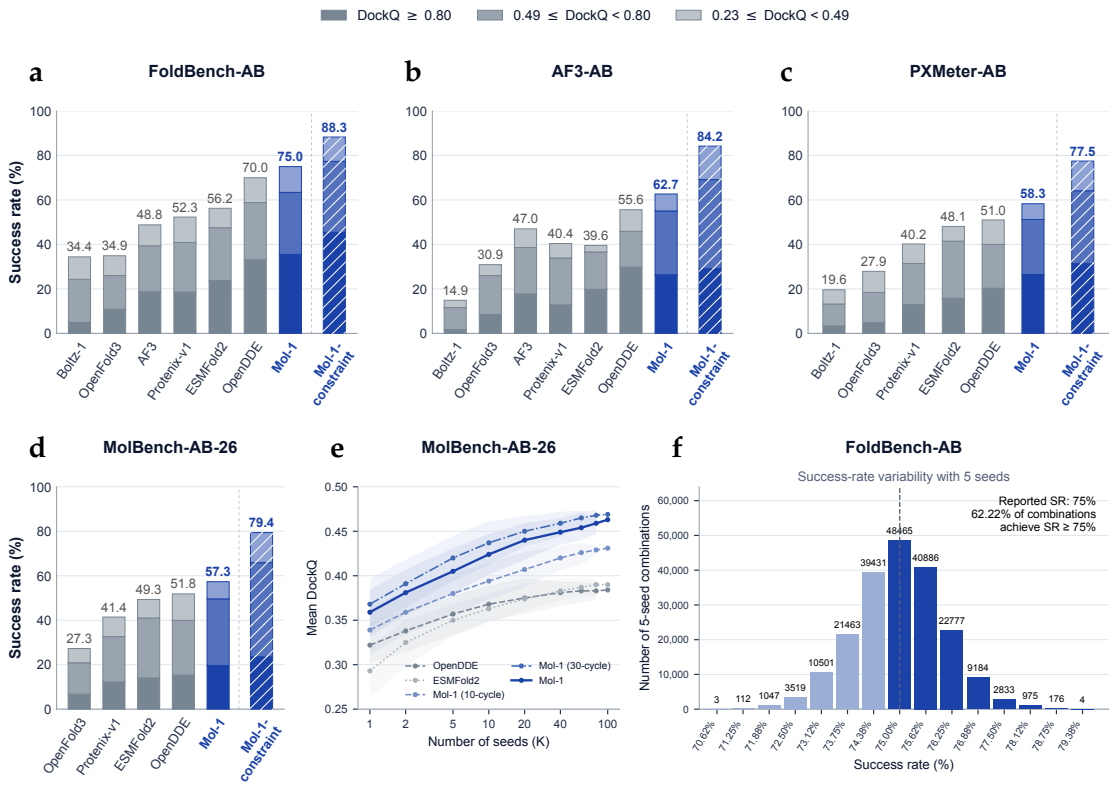


Figure 1 | Antibody–antigen structure prediction. **a–d,** Top-ranked DockQ success rates with a five-seed budget. **e,** Mean DockQ versus seed count K; shading denotes the 5th–95th percentiles over seed subsets. **f,** Success rates across all 201,376 five-seed subsets of 32 seeds on FoldBench-AB. Evaluation protocols and sources are detailed in Appendix A.2.

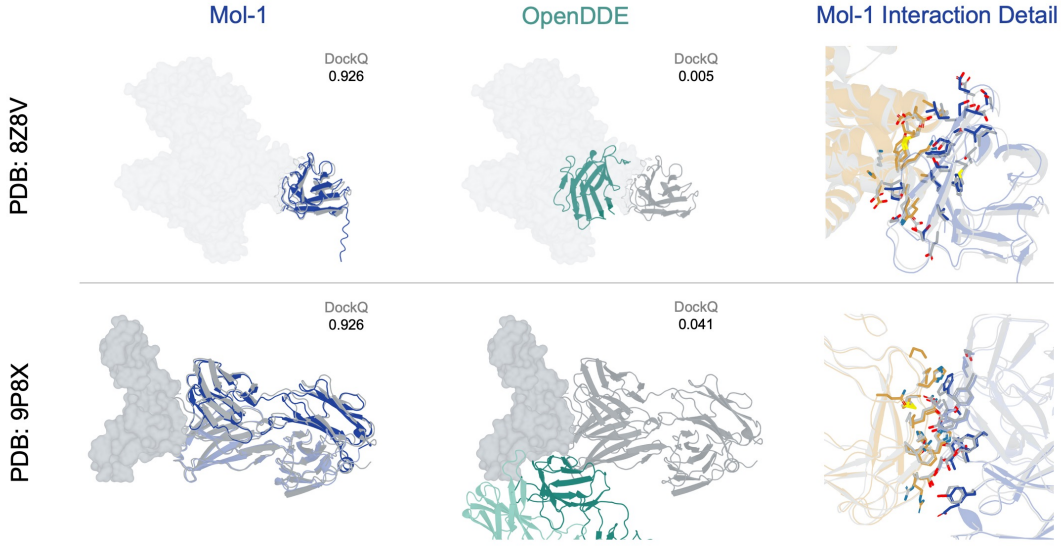


Figure 2 | Top-ranked antibody–antigen predictions: **a,** albumin–ALB8 VHH (PDB 8Z8V); **b,** GITR–Ab#1 Fab (PDB 9P8X). Mol-1 (blue, left) and OpenDDE (green, middle) are overlaid with experimental structures (gray). Right: enlarged Mol-1 interface overlays with selected residues shown as sticks. Antigens are surfaces; antibodies are cartoons. DockQ scores are shown above each comparison; GITR scores assess the antigen–heavy-chain interface.

20 cycles with 10 seeds, and 10 cycles with 40 seeds. With 5 samples per seed, these configurations require 150, 200, and 400 total trunk cycles (cycles × seeds), respectively. Thus, increasing recycling depth achieves comparable success rates under a smaller compute budget. However, gains from 20 to 30 cycles are smaller than those from 10 to 20 (Figure 1e), motivating our choice of 20 cycles to balance accuracy and computational cost for antibody–antigen prediction tasks. As recycling gains are less pronounced for Mol-1-constraint and for other molecular modalities, we retain 10 cycles in those settings for efficiency.

In antibody discovery, prior experiments or studies often provide information about the antigen binding region (epitope) even when the complex structure remains unknown. Mol-1-constraint allows researchers to incorporate this epitope information as constraints to guide structure prediction. With epitope input, it achieves success rates of 88.3%, 84.2%, 77.5%, and 79.4% at DockQ > 0.23 on FoldBench-AB, AF3-AB, PXMeter-AB, and MolBench-AB-26, respectively. This high accuracy supports its utility in epitope-informed antibody modeling, where recovering a plausible binding pose can help guide downstream experimental investigation.

Mol-1 enables accurate modeling of antibody–antigen complexes that have remained difficult to predict. We highlight two such cases spanning VHH and Fab formats: the albumin-binding ALB8 domain of ozoralizumab (Nanozora) (Mima et al., 2024) and a GITR-directed inhibitory Fab (Yan et al., 2026). Both interfaces are recovered with DockQ scores above 0.9, while none of the evaluated baseline models produces an acceptable top-ranked prediction. These cases involve interactions relevant to half-life extension and receptor signaling. Figure 2 compares Mol-1 and OpenDDE predictions and shows overlays of the Mol-1 predictions with the experimental structures. Scores correspond to the top-ranked prediction from each evaluated configuration. For the Fab complexes, DockQ evaluates the specified antigen–heavy-chain interface rather than the complete Fab–antigen interface.

Human serum albumin–ALB8 VHH (PDB 8Z8V). ALB8 is the albumin-binding VHH domain of ozoralizumab (Nanozora), in which it serves as a half-life-extension module (Mima et al., 2024). Mol-1 accurately reproduces the VHH binding position and orientation, achieving a DockQ of 0.926 and an interface RMSD of 0.521 Å, compared with DockQ of 0.005 for the OpenDDE baseline (Figure 2a). Mol-1 recovers 46 of the 51 native residue-pair contacts yielding a recall of 90.2%. These results show close agreement with the experimental structure in both the overall binding pose and the interfacial contact pattern.

GITR–Ab#1 Fab (PDB 9P8X). Ab#1 was investigated as a ligand-non-competitive inhibitor of GITR signaling, providing a Fab example relevant to antibody-mediated receptor regulation (Yan et al., 2026). Mol-1 achieves a 0.926 DockQ for the antigen–heavy-chain interface, compared with 0.041 for the OpenDDE baseline (Figure 2b). Within the scored interface, Mol-1 recovers all 18 native residue-pair contacts. In contrast, OpenDDE fails to reproduce the native orientation of the Fab relative to GITR.

2.2. Protein–Ligand Complexes

Predicting a protein–ligand complex requires accurately recovering both the ligand's binding pose and its interactions with the surrounding protein pocket. We evaluate Mol-1 on the Runs N' Poses (RNP) benchmark (Runs N' Poses contributors, 2025), where it outperforms all baselines across all eight similarity bins (Figure 3a). Notably, in the lowest-similarity bin, Mol-1 achieves a 36.1% success rate compared to 24.6% for AlphaFold 3. Incorporating binding-site information further boosts the success rate of Mol-1-constraint to 63.9% in this bin, yielding consistent improvements across all bins under the same inference budget.

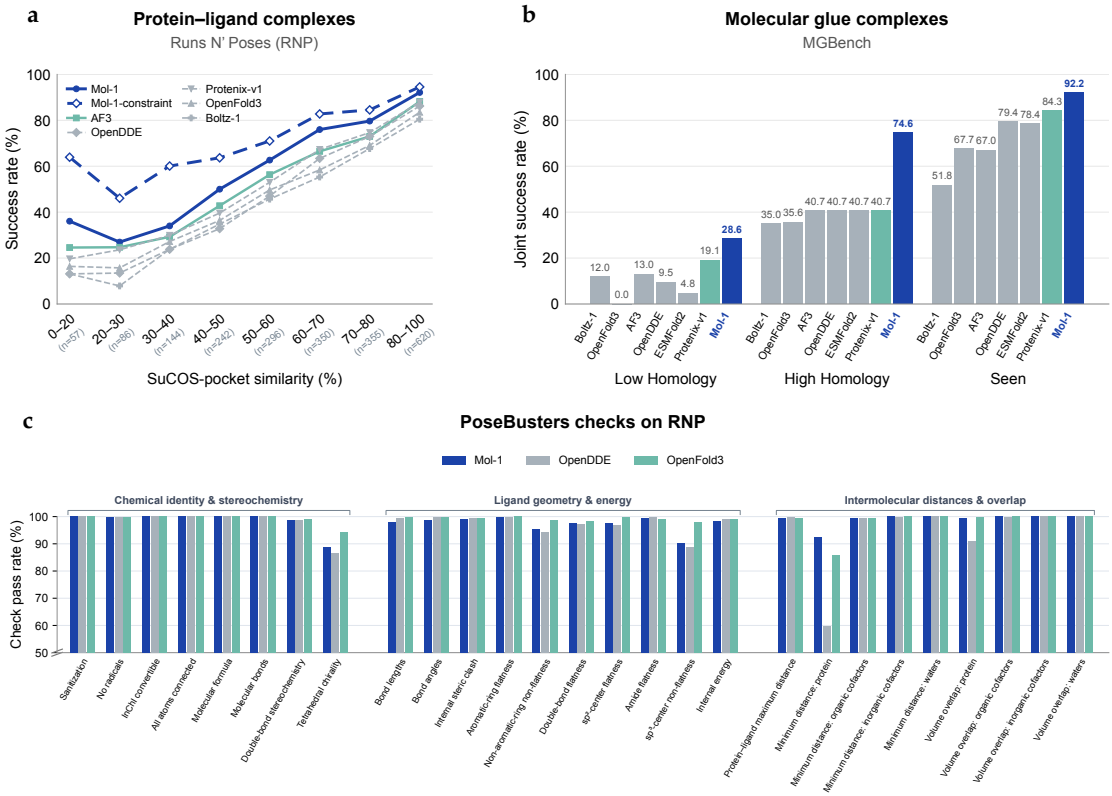


Figure 3 | Protein–ligand and molecular glue prediction. **a,** Runs N′ Poses success rates by SuCOS-pocket similarity to training complexes; parentheses give system counts. **b,** MGBench success rates by homology group. Success in **a** requires ligand RMSD < 2 Å and lDDT-PLI > 0.8, while success in **b** requires ligand RMSD < 2 Å, lDDT-PLI > 0.8 and protein-protein interface DockQ > 0.23. **c,** Individual PoseBusters check pass rates over 64,800 predictions per method. Definitions and evaluation details are provided in Appendix A.2.

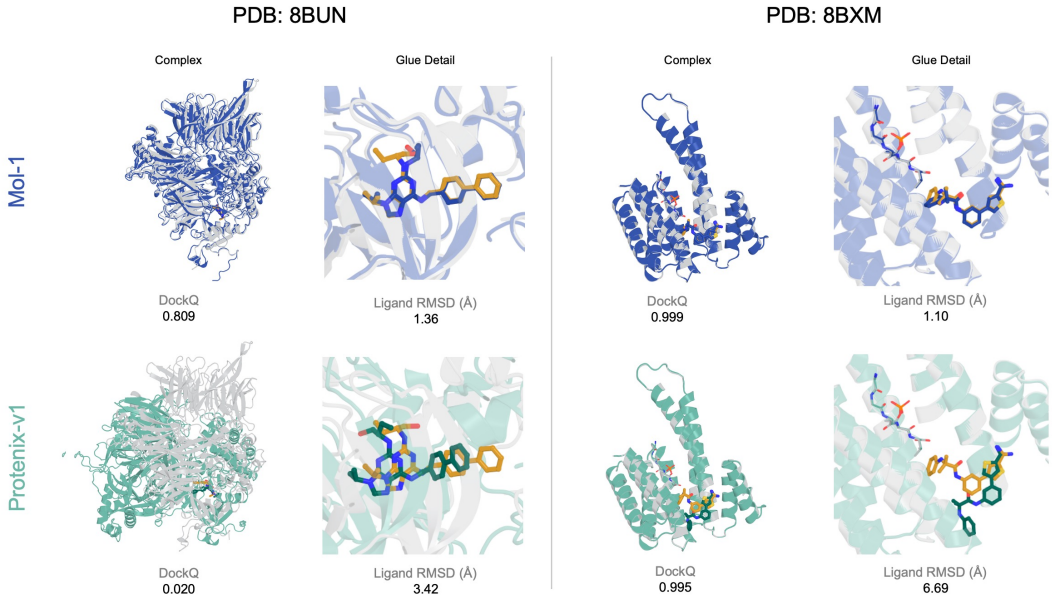


Figure 4 | Top-ranked molecular glue predictions: **a,** DDB1–CDK12–DS16 (8BUN); **b,** 14-3-3σ–ERα phosphopeptide–1074397 (8BXM). Mol-1 (top, blue) and Protenix-v1 (bottom, green) are overlaid with experimental structures (gray). Scoring details are provided in Appendix A.2.

Beyond pose accuracy, we assess the chemical and geometric validity of the predicted structures using PoseBusters (Buttenschoen et al., 2024). Mol-1 achieves high pass rates across individual PoseBusters criteria, although OpenFold 3 performs slightly better on ligand geometry and energy checks (Figure 3c). These evaluations complement the pose-recovery benchmarks by diagnosing stereochemical and geometric violations in the predicted complexes.

2.3. Molecular Glue Complexes

Molecular glues induce or stabilize protein–protein interactions, offering a route to therapeutic mechanisms such as targeted protein degradation (Konstantinidou and Arkin, 2024). Predicting their ternary complexes requires resolving both the protein–protein interface and the small molecule's contacts with its binding partners. We evaluate Mol-1 using the MGBench study's Seen, High Homology, and Low Homology groups, which distinguish earlier structures from held-out complexes with higher or lower sequence similarity to the study's training reference set (Liao et al., 2025).

Mol-1 achieves success rates of 92.2%, 74.6%, and 28.6% across these three groups, exceeding the displayed baselines in each (Figure 3b). On the Low Homology group, Mol-1 improves on the strongest displayed baseline from 19.1% to 28.6%. The lower success rates in this group show that predicting molecular-glue interactions in less familiar protein contexts remains challenging.

Accurate modeling of molecular-glue complexes requires recovering both the protein–protein arrangement and the ligand pose. We examine two cases that illustrate these distinct challenges and show how Mol-1 resolves them in its top-ranked predictions (Figure 4).

DDB1–CDK12–DS16 (PDB 8BUN). DS16 belongs to a class of molecular glues that recruit CDK12–cyclin K to DDB1, promoting cyclin K degradation (Kozicka et al., 2024). Mol-1 closely reproduces the experimental ternary architecture, achieving a DockQ of 0.809 for the DDB1–CDK12 interface and a ligand RMSD of 1.36 Å for DS16. Protenix-v1 shows some local agreement with the experimental ligand pose after pocket alignment (ligand RMSD of 3.42 Å), but substantially misplaces CDK12 relative to DDB1, yielding a DockQ of only 0.020. The agreement at the ligand pocket therefore does not extend to the overall ternary architecture. Mol-1 recovers both the relative placement of the proteins and the glue-binding pose, capturing how DS16 fits into the assembled protein interface.

14-3-3σ–ERα phosphopeptide–1074397 (PDB 8BXM). 1074397 is a fragment-linked, noncovalent stabilizer of the interface between 14-3-3σ and an ERα-derived phosphopeptide (Visser et al., 2023). Both Mol-1 and Protenix-v1 accurately predict the evaluated protein-protein interface, with DockQ scores of 0.999 and 0.995, respectively. Mol-1 also closely reproduces the experimental ligand position and orientation, achieving a ligand RMSD of 1.10 Å. Protenix-v1, however, misplaces the ligand with a ligand RMSD of 6.69 Å, illustrating that interface recovery alone does not guarantee correct placement of the molecular glue.

2.4. Protein–Peptide Complexes

We evaluate protein–peptide complex prediction on MolBench-Peptide-26, comprising 172 interface clusters from 150 PDB entries released in 2026 Q1–Q3. Mol-1 achieves success rates of 88.84%, 77.28%, and 54.07% at DockQ thresholds of > 0.23, > 0.49, and > 0.80, respectively, exceeding displayed baselines at each threshold (Figure 5). Binding-site conditioning further improves performance across all three thresholds, reaching 60.16% at DockQ > 0.80.

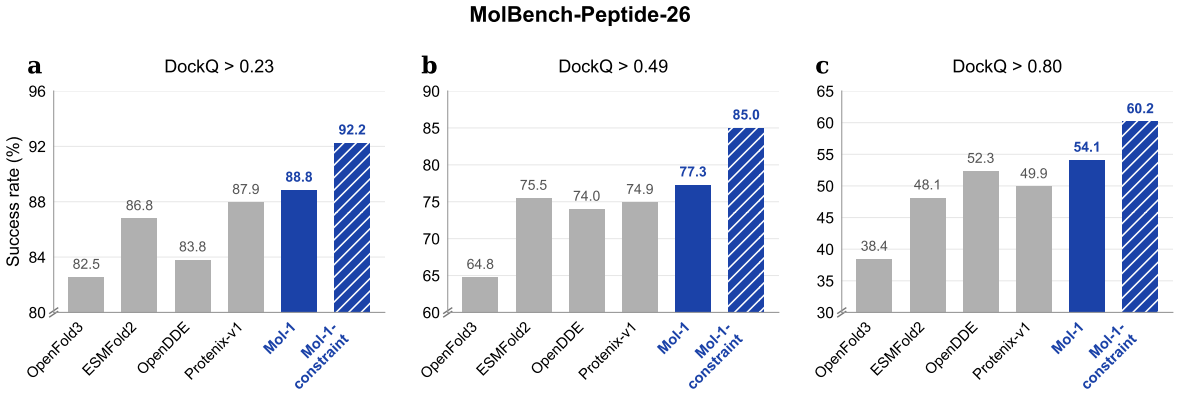


Figure 5 | Protein–peptide complex prediction on MolBench-Peptide-26, a 2026 Q1–Q3 benchmark comprising 172 interface clusters from 150 PDB entries. **a–c,** Success rates at DockQ > 0.23, > 0.49, and > 0.80, respectively. Mol-1 and Mol-1-constraint use 10 recycling cycles and are shown in solid and hatched blue; baselines are gray. Y-axes are truncated and use different ranges across panels.

3. Scaling Biomolecular Intelligence by Building Long-Running Agents

A prediction can be more than an answer: it can help determine what to investigate next.

Although biomolecular foundation models like Mol-1 are increasingly capable of predicting the structures of various types of complexes, structure prediction and assembly of very large protein complexes remain substantial challenges. Yet many large assemblies are also highly valuable and directly tied to therapeutics: the 20S proteasome and p97 are oncology targets, HBV capsid assembly is the target of capsid assembly modulators in clinical development (Janssen et al., 2023), and AAV capsids are delivery vehicles for approved gene therapies (Mendell et al., 2017; Russell et al., 2017). However, solving one experimentally by Cryo-EM can take an expert months to years, and some are too flexible or heterogeneous to resolve at all.

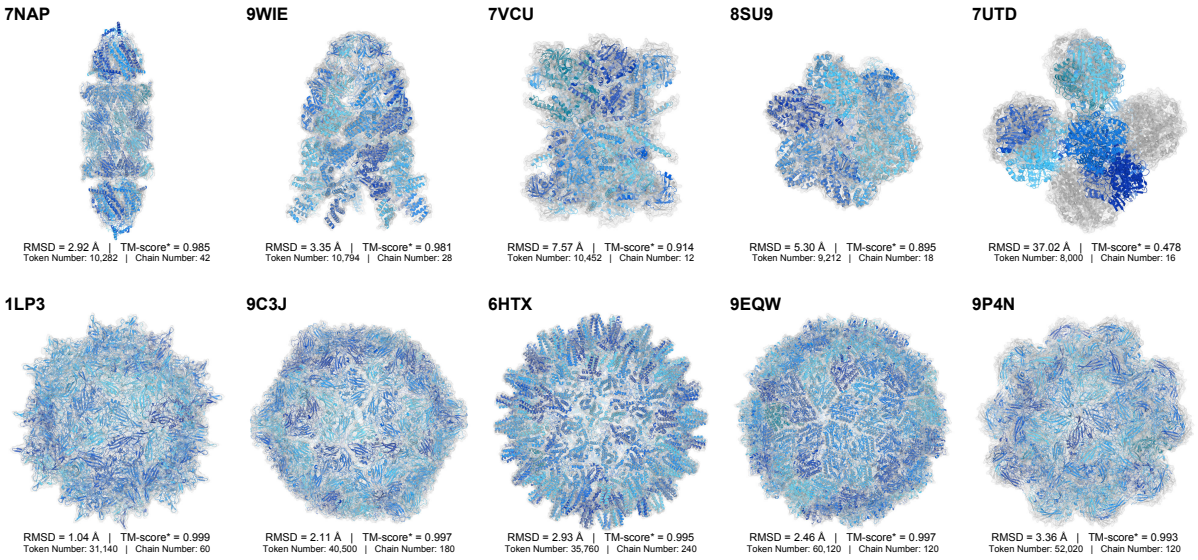


Figure 6 | Large-complex structure prediction with the Mol-1 agent. Ten assemblies are identified by PDB ID; each panel reports RMSD, TM-score, token count, and chain count.

3.1. The Scaling of Biomolecular Structure Prediction

The largest complex actually predicted by AlphaFold 3 was reported to be the 40S ribosomal subunit, 7,663 residues. Anthropic's Big mode then carried a set of molecular machines from 4k to about 11k tokens onto an 8-GPU node. The largest accurately predicted complex contained 10,761 tokens, marking the reported limit of accurate prediction in that study. At capsid scale — seven complete viral capsids and protein shells of roughly 31k to 70k residues, on a node of eight B300 GPUs — every prediction collapsed (Claude Science et al., 2026).

On top of Mol-1, we built a long-running agent that predicts the structures of complex assemblies up to 60,000 tokens, 6× beyond where previous approaches achieved, running for tens of hours on a single NVIDIA H20 GPU. As shown in Figure 6, 8 out of 10 assemblies were accurately predicted with a TM-score (Zhang, 2008) of 0.90 or higher, and predictions for 9 out of 10 assemblies outperformed previously reported results (Claude Science et al., 2026) by a substantial margin. The accurately predicted assemblies included five viral capsids for which previous methods had produced collapsed structures.

3.2. How the Agent Works

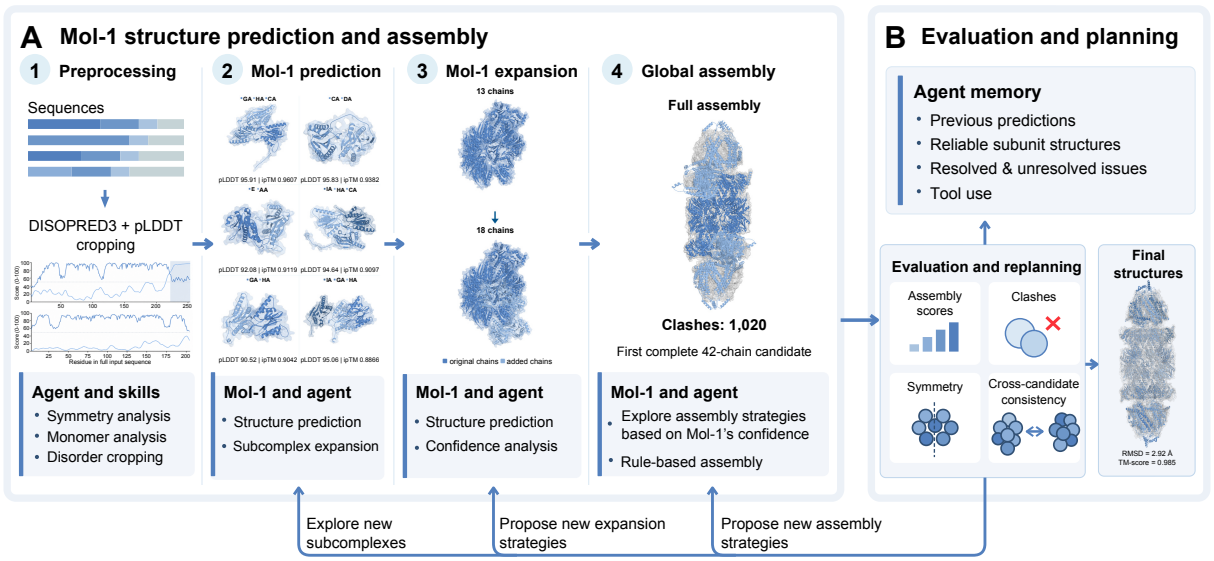


Figure 7 | **Long-running agent workflow for iterative protein complex prediction and assembly. A,** DISOPRED3 and prediction confidence guide sequence preprocessing, followed by Mol-1 sub-complex prediction, expansion and global assembly. **B,** During extended execution, the agent repeatedly evaluates candidate assemblies using assembly scores, clashes, symmetry and consistency across candidates. Evaluation guides successive rounds of replanning, prediction and refinement. Agent memory carries previous predictions, reliable subunit structures and unresolved issues across rounds, supporting sustained execution and iterative improvement.

Why is it challenging to predict these large assemblies? First, their token counts often far exceed the crop sizes used during model training, making it difficult for models to capture global topological relationships and generate plausible overall conformations, potentially leading to structural collapse. Second, for methods that assemble complexes from local structures, the candidate assembly space grows combinatorially with increasing chain and token counts. Furthermore, errors due to imperfect local predictions can accumulate as the assembly grows.

To address these challenges, we developed an agent-driven multistage framework for struc-

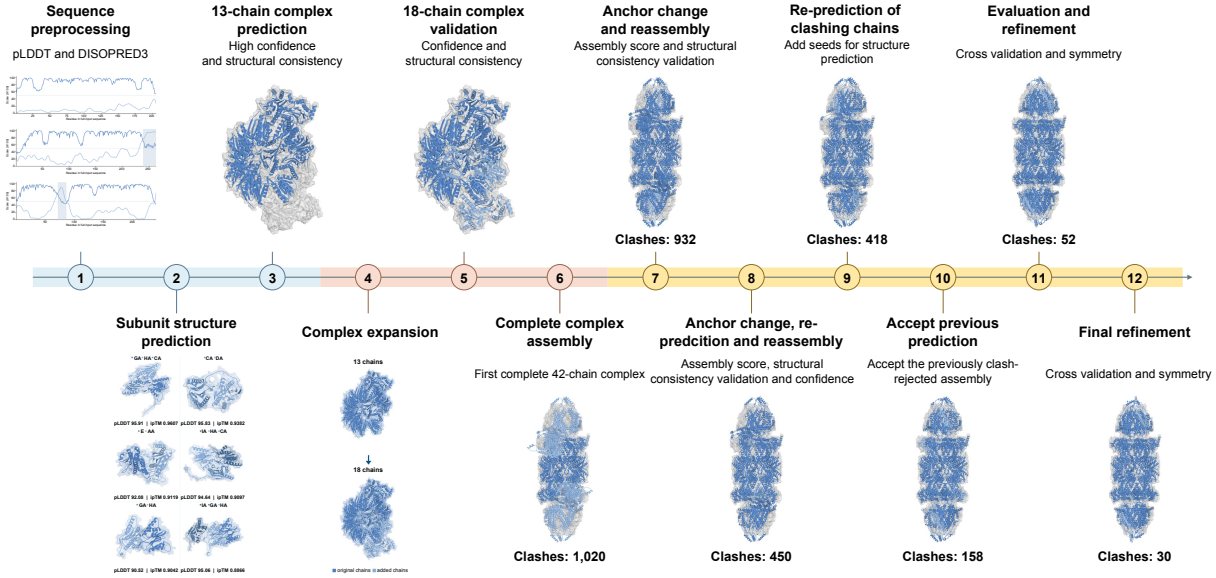


Figure 8 | **Key steps in agent-guided prediction and refinement of 7NAP.** Numbered stages highlight selected key steps across the agent's full execution trajectory. These span sequence pre-processing, subunit prediction, expansion from a 13-chain to an 18-chain complex and assembly of the complete 42-chain complex. Later steps include anchor changes, re-prediction, reassembly and reconsideration of a previously rejected assembly, guided by confidence, structural consistency and symmetry. Clash counts are shown beneath the selected assemblies, from 1,020 in the first complete candidate to 30 after refinement.

ture prediction and assembly that integrates the high-accuracy structure prediction capabilities of Mol-1, the established expertise in structure prediction (Liu et al., 2026; Shor and Schneidman-Duhovny, 2024; Wang et al., 2026), and the task planning, hypothesis generation, and strategy validation capabilities of LLMs. We also incorporated additional objective evaluation metrics into the iterative loop, including cross-candidate structure consistency and backbone steric clash assessment (Figure 7). Moreover, we illustrate the key steps in agent-guided prediction and refinement using 7NAP as a case study (Figure 8).

This framework has two modules:

- **Generation and Assembly of Structures.** Monomer and symmetry analyses, combined with DISOPRED3 (Jones and Cozzetto, 2015) and pLDDT scores, guide the identification and removal of disordered or low-confidence regions during assembly to avoid clashes. The agent predicts local subcomplexes, selects reliable combinations using confidence scores, and expands them into progressively larger subcomplexes. Then, it generates 16 full-complex candidates under different assembly strategies.
- **Evaluation and Planning.** The agent scores each candidate on assembly confidence, steric clashes, symmetry, and cross-candidate consistency, retaining previous predictions, reliable substructures, identified problems, and tool-use history in memory. Guided by these scores, it revises its strategy for subcomplex exploration, expansion, and global assembly, looping through prediction, evaluation, and replanning until the stopping criteria are met. Difficult assemblies are conventionally handled with hand-tuned heuristics; here the choice of tool and strategy is driven by sequence features and evaluation feedback, and hard cases simply receive a longer iteration budget.

4. Design Meets Experiment

We developed a *de novo* VHH design workflow using an earlier checkpoint of Mol-1, without training an additional diffusion model specifically for binder generation. This approach builds on the predictive model's capabilities, complementing methods based on dedicated generative models (Bennett et al., 2026; Stark et al., 2025). We applied this workflow to *de novo* VHH design across drug discovery campaigns and systematically evaluated the resulting designs for binding affinity, physicochemical properties, and functional activity in cell-based assays.

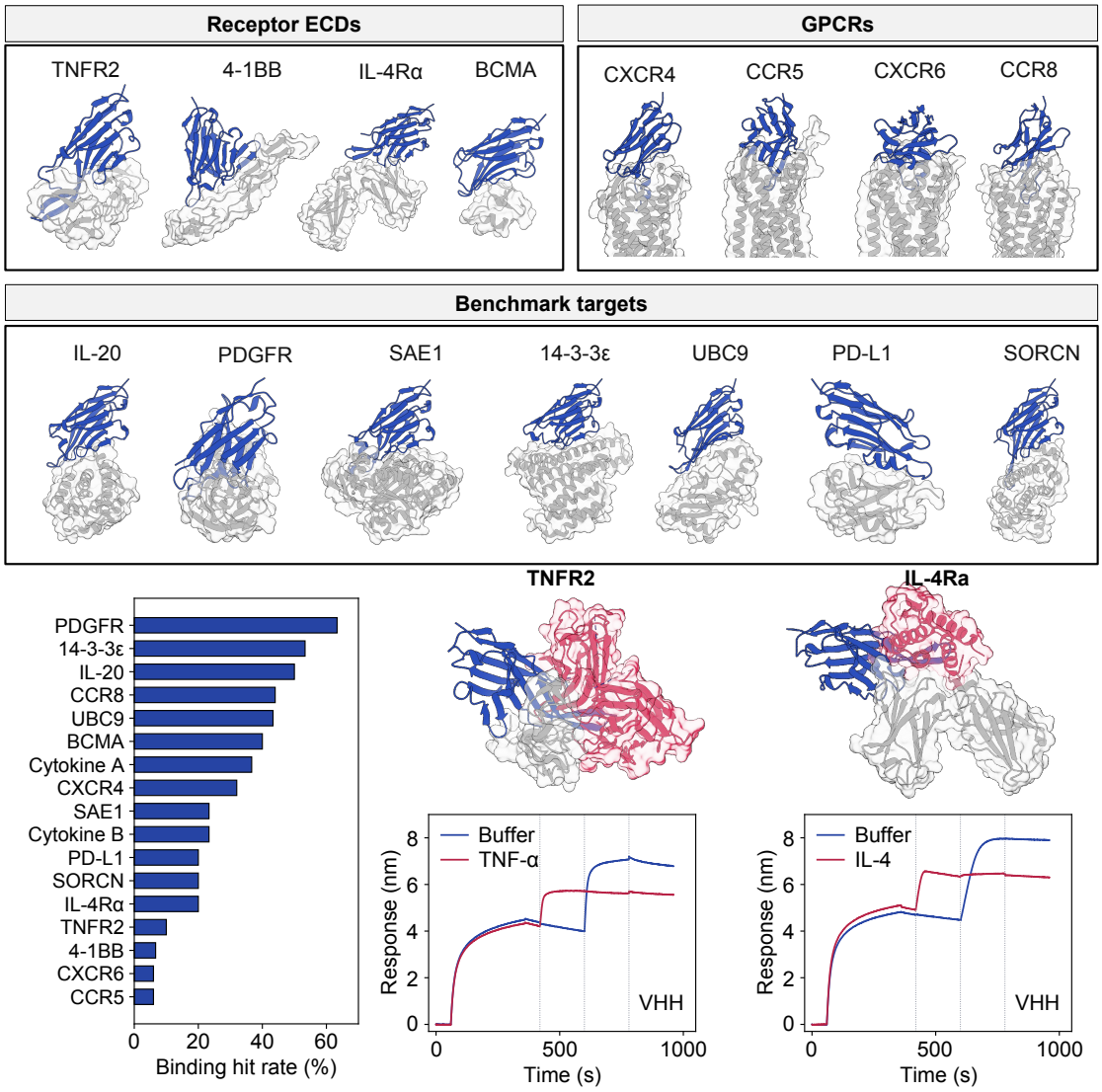


Figure 9 | **Target coverage, binding-hit rates, and ligand competition.** The upper two rows show designed VHH–target complex models grouped as receptor extracellular domains, GPCRs, and benchmark targets. The lower-left chart shows binding-hit rates across 17 targets (Table 3). The lower-middle and lower-right panels show receptor-aligned structural overlays and sequential BLI curves for TNFR2 and IL-4Rα, respectively. Overlays compare the designed VHH and natural ligand binding sites. The red curves represent receptor-loaded sensors exposed to TNF-α or IL-4, while the blue curves represent sensors exposed to assay buffer.

4.1. Binding across Targets and Recognition of Designed Epitopes

De novo-designed VHHs yielded binding hits across 17 targets, ranging from soluble proteins, receptor extracellular domains to GPCRs, with initial hit rates ranging from 6% to 63% (Figure 9). To assess the ability to target predefined epitopes, we designed VHHs against the natural ligand-binding regions of TNFR2 and IL-4Rα and evaluated their binding sites using sequential biolayer interferometry (BLI) competition assays. Prebinding of TNF-α to TNFR2 reduced subsequent binding of TNFR2 VHH 01, while prebinding of IL-4 to IL-4Rα reduced binding of IL-4Rα VHH 01 (Figure 9). These results support binding to the intended ligand-binding regions, demonstrating the potential of *de novo* VHH design for epitope-directed targeting. Epitope-directed design could streamline therapeutic binder discovery by reducing the need for downstream epitope screening. Binding hits from 13 of the 17 targets yielded sufficient quantities of protein for further biophysical characterization.

4.2. Developability of Designed VHHs

For 12 of 13 targets (92.3%), we identified candidates that passed at least three of the four biophysical screening criteria. Overall, 23 candidates met this criterion (Figure 10a). The panel assessed hydrophobicity, thermal stability, self-interaction, and nonspecific binding using HIC, DSF, AC-SINS, and BVP assays, respectively. Twenty-four candidates (80.0%) met the HIC criterion, and 24 (80.0%) had a reported T_{m1} above 60 °C. Twenty-five candidates (83.3%) showed AC-SINS wavelength shifts below 10 nm, without a high self-interaction signal under the applied screening definition (Figure 10b–e). Together, these results demonstrate that designed VHH-Fc candidates combined favorable biophysical properties across diverse targets.

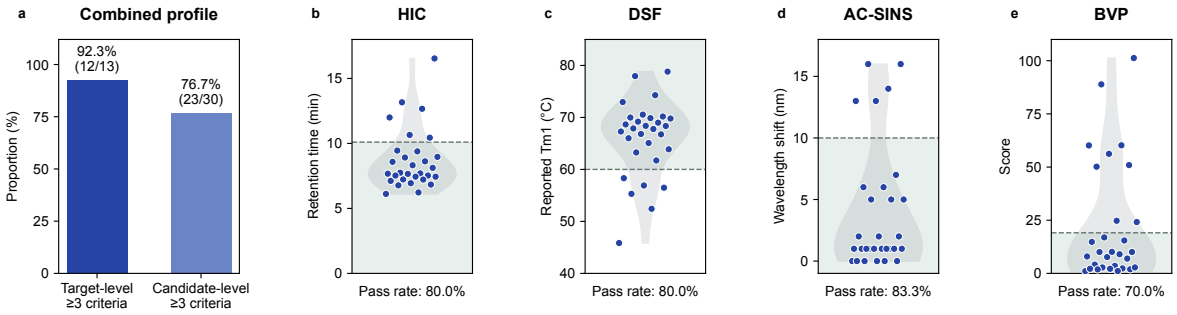


Figure 10 | **Biophysical screening of designed VHH-Fc candidates across targets.** Thirty candidates across 13 targets had complete measurements for all four assays. **a**, Proportions of targets with ≥ 1 candidate meeting ≥ 3 of four criteria (left) and candidates meeting ≥ 3 criteria (right). **b–e**, HIC retention time, reported DSF T_{m1}, AC-SINS wavelength shift and BVP score, respectively. Dots represent individual candidates. Green shading and dashed lines indicate screening ranges and thresholds; pass rates appear below each assay panel.

4.3. Potent Binding and Cellular Activity

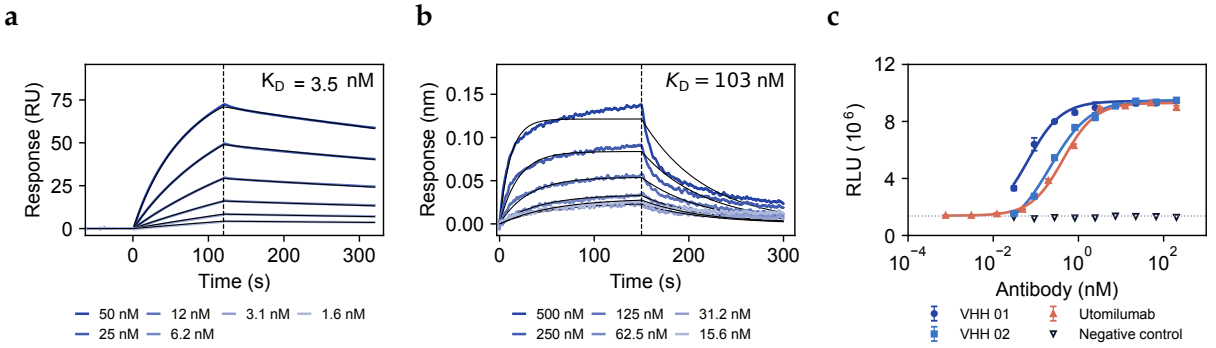
4.3.1. Potent 4-1BB Agonistic VHHs with Favorable Developability

4-1BB (CD137) is a costimulatory receptor that promotes T-cell activation, proliferation, and survival, making it an attractive target for cancer immunotherapy (Singh et al., 2024). Tumor-

targeted and bispecific 4-1BB agonists offer opportunities to incorporate receptor costimulation into multifunctional therapeutic architectures (Klein et al., 2024).

Mol-1-designed VHHs combined potent 4-1BB agonism with favorable developability. VHH 01 bound 4-1BB with an SPR K_D of 3.5 nM, and multi-concentration BLI confirmed binding by VHH 02 (Figure 11a,b). In a six-hour reporter assay, VHH 01 and VHH 02 achieved activation EC_{50} values of 0.05 and 0.22 nM, respectively, compared with 0.43 nM for Utomilumab (Figure 11c). Beyond their potent agonistic activity, both VHHs demonstrated favorable developability profiles, characterized by high monomeric purity, good hydrophilicity, high thermal stability, and low propensity for self-interaction (Figure 11d).

Together, their potent agonistic activity and favorable biophysical profiles highlight these VHHs as compact modules for engineering bispecific and multispecific antibody therapeutics.



d

Candidate	Activation EC_{50} (nM)	SEC purity (280 nm, %)	HIC RT (min)	DSF T_{m1} (°C)	AC-SINS Δλ (nm)
4-1BB VHH 01	0.05	100	7.44	70.5	0
4-1BB VHH 02	0.22	97	6.23	70.2	1

Figure 11 | **4-1BB binding kinetics, agonist dose responses, and molecular profiles. a,b**, SPR sensorgrams for VHH 01 and BLI responses for VHH 02, respectively, measured as VHH-Fc fusions; black lines show fitted curves. For VHH 02, the six concentrations were globally fitted with a 1:1 binding model. **c**, Six-hour reporter dose responses of VHH 01 (dark blue), 02 (light blue), Utomilumab (orange, the positive control), and a VHH-Fc negative control (grey). **d**, Activity and molecular profiles of the two designs.

4.3.2. Lab in the Loop

One round of experimental feedback increased the 4-1BB functional hit rate from 7% (2/30) to 85% (51/60) (Figure 12a). Both first-round binding hits were functional agonists, and their experimental results guided the second round of design. At 0.03 nM, 48 of the 51 second-round agonists exceeded the mean activation of Utomilumab (Figure 12b). Experimental feedback thus expanded the pool of functional agonists while retaining strong activation at low concentration.

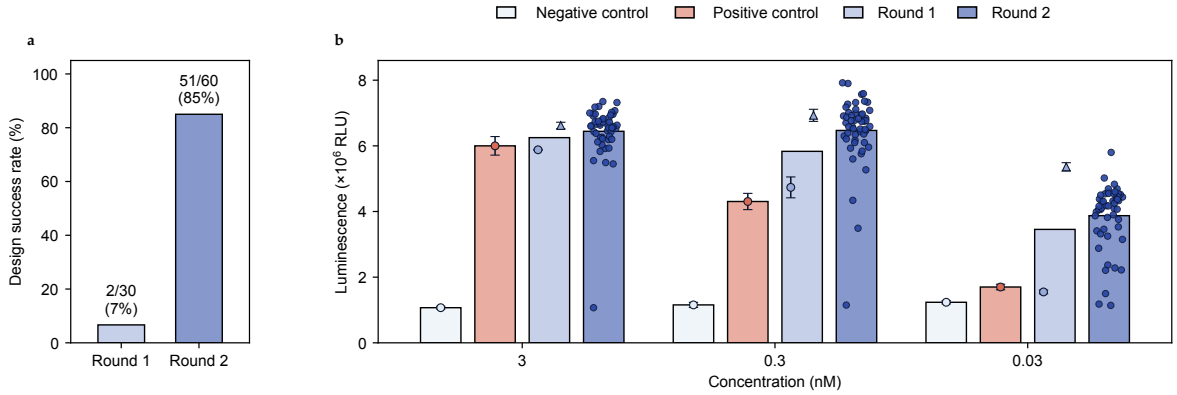


Figure 12 | **4-1BB functional hit rates and activation across two design rounds. a**, Functional hit rates increased from 2/30 (7%) in Round 1 to 51/60 (85%) in Round 2. **b**, Luminescence responses of two Round 1 designs, 51 Round 2 designs, and positive and negative controls at three concentrations. At 0.03 nM, 48 of 51 Round 2 designs exceeded the positive-control mean. The negative control is an unrelated VHH-Fc construct.

4.3.3. BCMA: Zero-Shot Design Achieves 43 pM Affinity

BCMA-targeted T-cell engagers and CAR-T therapies have established BCMA as a therapeutic target in multiple myeloma. BCMA-binding affinity is a key engineering parameter in T-cell engager design (Chen et al., 2021), while the use of two BCMA-binding VHH domains in cilta-cel demonstrates the therapeutic utility of compact single-domain recognition modules (Banerjee et al., 2022). Mol-1 yielded 12 binding hits from 30 zero-shot BCMA designs. VHH 01 and VHH 02 achieved SPR-derived K_D values of 43 pM and 1.1 nM, respectively (Figure 13a,b). Both also showed strong binding to cell-surface BCMA, with fitted cell-binding EC_{50} values of 0.19 and 0.16 nM, compared with 4.1 nM for the belantamab biosimilar (Figure 13c), corresponding to 22- and 26-fold lower EC_{50} values. These results establish computationally designed VHHs as high-affinity BCMA recognition modules and highlight their compact single-domain architecture as a foundation for engineering multivalent and multispecific antibody therapeutics.

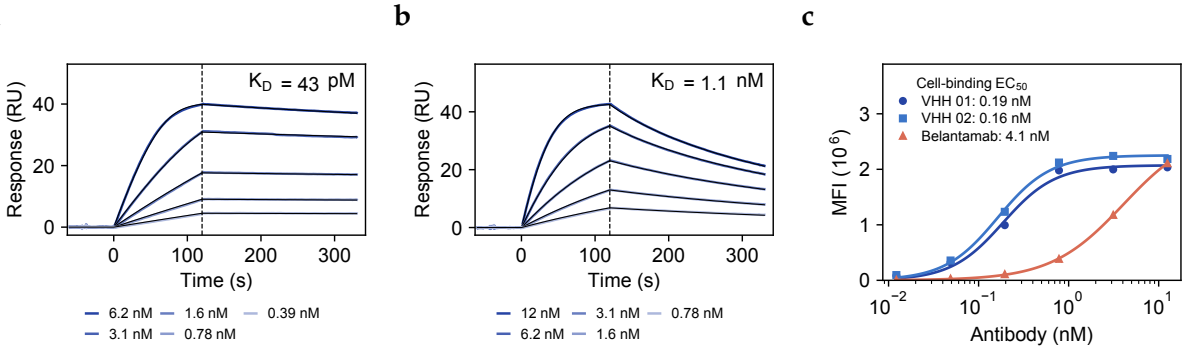


Figure 13 | **BCMA binding kinetics and cell-binding dose responses. a,b**, SPR sensorgrams for BCMA VHH 01 and VHH 02, respectively; black curves show fits, with K_D values of 43 pM and 1.1 nM. **c**, Binding to human BCMA-expressing cells by VHH 01 (dark blue circles), VHH 02 (light blue squares), and Belantamab (orange-red triangles). Symbols show measured median fluorescence intensity (MFI), and solid lines show fitted curves. The fitted cell-binding EC_{50} values were 0.19, 0.16, and 4.1 nM, respectively, corresponding to approximately 20-fold lower EC_{50} values for both designed VHHs than for Belantamab.

4.3.4. *De Novo VHH Design Enables CXCR6 Binding and Functional Antagonism*

CXCR6 is a seven-transmembrane chemokine receptor that binds CXCL16 and regulates immune cell trafficking and localization. The CXCL16–CXCR6 signaling axis also plays an important role in positioning and sustaining cytotoxic T cells within the tumor microenvironment. CXCR6 VHH 01 combined cell-surface binding with functional antagonism of CXCL16-induced signaling. The designed VHH bound to a polyclonal population of HEK293T cells stably overexpressing CXCR6, with a cell-binding EC_{50} of 39 nM (Figure 14a). In a cAMP assay, 30 nM CXCL16 reduced the mean response to 70.6% of the forskolin-only control. Addition of 167 nM VHH restored the response to 92.6% (Figure 14b). These results demonstrate both recognition and functional modulation of a seven-transmembrane chemokine receptor by a designed VHH.

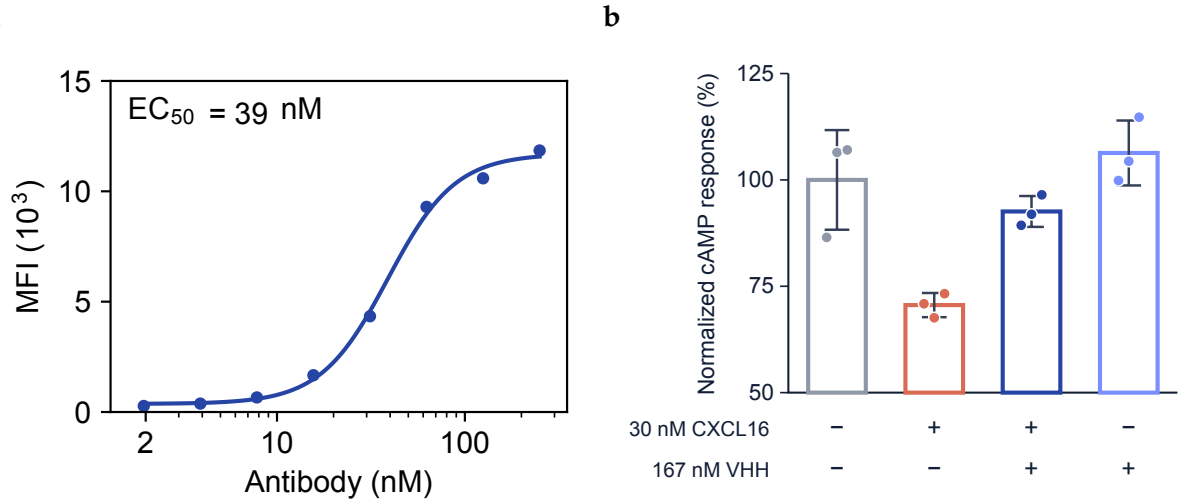


Figure 14 | **CXCR6 cell-binding dose response and cAMP signaling. a**, CXCR6 VHH 01 cell binding across eight concentrations; the 4PL fit yielded a cell-binding EC_{50} of 39 nM. **b**, cAMP responses under four conditions: forskolin-only control, 30 nM CXCL16, 30 nM CXCL16 with 167 nM VHH, and 167 nM VHH alone, from left to right. Treatment rows indicate presence (+) or absence (−). Responses were normalized to the mean forskolin-only control response and expressed as percentages. Bars and error bars show means ± sample SD of three readings per condition; dots show individual readings. Addition of VHH restored the mean response to 92.6% of the forskolin-only control.

4.4. Design VHHs to Bind Two Antigens

The ability to recognize multiple related antigens is a desirable property for antibodies, but achieving such cross-reactivity through conventional screening can be challenging. To investigate whether cross-reactivity could be explicitly programmed into *de novo* antibody design, we designed VHHs to recognize both cytokine-A and cytokine-B, two non-orthologous protein antigens sharing only 54% ungapped sequence identity. Five of 30 designs bound both antigens (Figure 15a). The same set yielded 11 hits against cytokine-A and 7 against cytokine-B (Figure 9). Across the five dual-binding candidates, the fitted $K_{D,1}$ values ranged from 14 to 251 nM for cytokine-A and from 2.2 to 95 nM for cytokine-B. These values represent the lower component dissociation constants from a global 2:1 fit to six-concentration BLI data. VHH 01 bound both antigens across the concentration series (Figure 15b,c). These findings demonstrate the feasibility

of explicitly designing for cross-reactivity and suggest a potential strategy for reducing the screening effort required to identify dual-antigen binders.

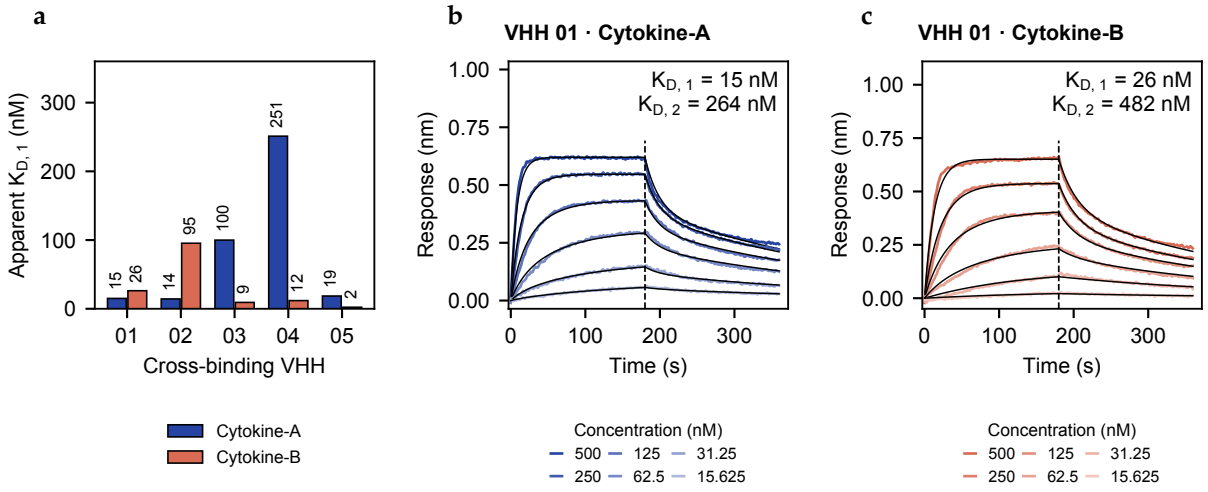


Figure 15 | **Binding affinities and BLI responses of dual-antigen VHHs.** Five VHH-Fc candidates bound both cytokine-A and cytokine-B. All candidates were analyzed at six concentrations (15.625–500 nM) and fitted with the same global 2:1 model. **a**, Fitted $K_{D,1}$ values for cytokine-A (blue) and cytokine-B (coral), spanning 14–251 nM and 2.2–95 nM, respectively. $K_{D,1}$ and $K_{D,2}$ denote the lower and higher component dissociation constants. **b,c**, VHH 01 BLI responses against cytokine-A and cytokine-B, respectively. Darker traces indicate higher concentrations; black lines show fits, and both component constants are labeled.

5. Discussion

The studies presented here examine how biomolecular prediction can support a broader range of scientific tasks. Mol-1 provides a common foundation for modeling interactions across molecular types, while task-specific workflows extend what can be achieved with that foundation.

The VHH studies connect target coverage to defined recognition objectives and useful molecular activities. Initial screens yielded binding hits across soluble proteins, single-pass receptors, and GPCRs, while ligand-competition experiments examined an intended epitope of binding. BCMA cell recognition and 4-1BB reporter activation connect biochemical binding to cellular readouts. CXCR6 cell recognition and the single-concentration cAMP response consistent with antagonism, together with cross-binding to cytokine-A/B extend the range of targets and recognition objectives. Broader biophysical measurements further characterize candidates for development.

These results have limitations. First, like existing structure prediction models, Mol-1 still produces steric clashes when modeling complex interfaces or large molecular assemblies. Second, the antibody design campaign used early Mol-1 checkpoints. The substantial gains in design success rate and potency following a single round of experimental feedback suggest that further improvements may be achievable through continued model development and iterative experimental validation.

We release Mol-1 not as a finished account of what biomolecular models can do, but as a

foundation for what others may build. With Mol-1, we aim to push the frontiers of biomolecular intelligence for the benefit of all, bringing us closer to a future where every disease can be cured.

Acknowledgments

Molence Bio is new; many of the relationships that made it possible are not. We are grateful to our colleagues, collaborators, and friends across the field—for the ideas they shared, the questions they asked, and the generosity with which they helped this work take shape.

With this first release, we look forward to the conversations, collaborations, and discoveries that may follow.

October 2026
When a shared belief becomes a beginning.

References

J. Abramson, J. Adler, J. Dunger, et al. Accurate structure prediction of biomolecular interactions with AlphaFold 3. *Nature*, 630:493–500, 2024. doi: 10.1038/s41586-024-07487-w.

M. Bailly, C. Mieczkowski, V. Juan, E. Metwally, D. Tomazela, J. Baker, M. Uchida, E. Kofman, F. Raoufi, S. Motlagh, Y. Yu, J. Park, S. Raghava, J. Welsh, M. Rauscher, G. Raghunathan, M. Hsieh, Y.-L. Chen, H. T. Nguyen, N. Nguyen, D. Cipriano, and L. Fayadat-Dilman. Predicting antibody developability profiles through early stage discovery screening. *mAbs*, 12(1): 1743053, 2020. doi: 10.1080/19420862.2020.1743053.

R. Banerjee, S. S. Lee, and A. J. Cowan. Innovation in BCMA CAR-T therapy: Building beyond the Model T. *Frontiers in Oncology*, 12:1070353, 2022. doi: 10.3389/fonc.2022.1070353.

N. R. Bennett, J. L. Watson, R. J. Ragotte, A. J. Borst, D. L. See, C. Weidle, R. Biswas, Y. Yu, E. L. Shrock, R. Ault, et al. Atomically accurate de novo design of antibodies with rfdiffusion. *Nature*, 649(8095):183–193, 2026.

M. Biasini, T. Schmidt, S. Bienert, V. Mariani, G. Studer, J. Haas, N. Johner, A. D. Schenk, A. Philippsen, and T. Schwede. OpenStructure: an integrated software framework for computational structural biology. *Acta Crystallographica Section D: Biological Crystallography*, 69(5): 701–709, 2013. doi: 10.1107/S0907444913007051.

D. P. Buckley, T. J. Vickers, A. Ellebedy, Z. T. Berndsen, and J. M. Fleckenstein. Bps2025-human infection with enterotoxigenic e. coli elicits multiple modes of antibody-mediated inhibition of the eata mucin-degrading protease. *Biophysical Journal*, 124(3):48a, 2025.

D. P. Buckley, M. Akhtar, M. Thapa, A. Schmitz, J. Turner, T. J. Vickers, N. Khatoon, M. H. Kaisar, J. A. Coggin, D. Ganguli, et al. Human enterotoxigenic escherichia coli (etec) infections elicit antibodies that broadly neutralize mucinases of pathogenic escherichia coli and shigella. *Proceedings of the National Academy of Sciences*, 123(25):e2614012123, 2026.

M. Buttenschoen, G. M. Morris, and C. M. Deane. PoseBusters: AI-based docking methods fail to generate physically valid poses or generalise to novel sequences. *Chemical Science*, 15: 3130–3139, 2024. doi: 10.1039/D3SC04185A.

ByteDance AML AI4Science Team, X. Chen, Y. Zhang, et al. Protenix—advancing structure prediction through a comprehensive AlphaFold3 reproduction. *bioRxiv*, 2025. doi: 10.1101/2025.01.08.631967.

S. Candido, T. Hayes, A. Derry, R. Rao, Z. Lin, R. Verkuil, B. Z. Wu, J. S. Lee, E. S. Bruguera, J. A. Keval, M. Kopylov, J. E. Pak, W. Wu, N. Thomas, S. Mataraso, A. Hsu, A. C. Trotman-Grant, K. Fatras, A. dos Santos Costa, R. Badkundri, H. Akın, D. Oktay, J. Deaton, E. Montabana, H. Sitwala, Y. Yu, M. Wiggert, D. A. Carlin, A. W. Goering, T. Blazejewski, M. Sandora, M. Hla, T. Z. Jia, L. H. Kloker, N. J. Sofroniew, M. Uehara, J. Pannu, S. Bachas, D. S. Liu, T. Sercu, and A. Rives. Language modeling materializes a world model of protein biology. *bioRxiv*, 2026. doi: 10.64898/2026.06.03.729735. URL https://www.biorxiv.org/content/10.64898/2026.06.03.729735. Preprint.

Chai Discovery Team. Chai-1: Decoding the molecular interactions of life. *bioRxiv*, 2024. doi: 10.1101/2024.10.10.615955.

Chai Discovery Team. Zero-shot antibody design in a 24-well plate. *bioRxiv*, 2025a. doi: 10.1101/2025.07.05.663018.

Chai Discovery Team. Drug-like antibody design against challenging targets with atomic precision. *bioRxiv*, 2025b. doi: 10.1101/2025.11.29.691346.

W. Chen, F. Yang, C. Wang, J. Narula, E. Pascua, I. Ni, S. Ding, X. Deng, M. L.-H. Chu, A. Pham, X. Jiang, K. C. Lindquist, P. J. Doonan, T. Van Blarcom, Y. A. Yeung, and J. Chaparro-Riggers. One size does not fit all: navigating the multi-dimensional space to optimize t-cell engaging protein therapeutics. *mAbs*, 13(1):1871171, 2021. doi: 10.1080/19420862.2020.1871171.

Claude Science, R. Shuai, R. Badkundri, V. Fan, K. Fatras, L. Jarosch, and A. Shanehsazzadeh. Accelerating open-source biomolecular models with Claude. Technical report, Anthropic, Sept. 2026. URL https://www-cdn.anthropic.com/e96b5807039a88168733d9687afe41dfbbd5de13.pdf. Released September 17, 2026; updated September 21, 2026.

K. Didi, Z. Zhang, G. Zhou, D. Reidenbach, Z. Cao, S. Cha, T. Geffner, C. Dallago, J. Tang, M. Bronstein, et al. Scaling atomistic protein binder design with generative pretraining and test-time compute. In *International Conference on Learning Representations*, volume 2026, pages 61056–61109, 2026.

J. M. Dodson, C. G. Clark, L. A. Lockhart, B. M. Leo, J. W. Schroeder, and B. J. Mann. Comparison of adherence, cytotoxicity, and gal/galnac lectin gene structure in entamoeba histolytica and entamoeba dispar. *Parasitology International*, 46(3):225–235, 1997.

C. Dominguez, R. Boelens, and A. M. Bonvin. Haddock: a protein- protein docking approach based on biochemical or biophysical information. *Journal of the American Chemical Society*, 125 (7):1731–1737, 2003.

J. Geiping, S. McLeish, N. Jain, J. Kirchenbauer, S. Singh, B. Bartoldson, B. Kailkhura, A. Bhatele, and T. Goldstein. Scaling up test-time compute with latent reasoning: A recurrent depth approach. In *Advances in Neural Information Processing Systems*, volume 38, 2025. URL https://proceedings.neurips.cc/paper_files/paper/2025/hash/3b01972cf31e6fa0fe29e4b8b5c2a0a1-Abstract-Conference.html.

S. F. Gérard, C. Redfield, and M. K. Higgins. Structural basis for carbohydrate recognition by the gal/galnac lectin of entamoeba histolytica involved in host cell adhesion. *PLoS Pathogens*, 22(2):e1013948, 2026.

Isomorphic Labs Team. Accurate predictions of novel biomolecular interactions with IsoDDE. Technical report, February 2026. URL https://storage.googleapis.com/isomorphiclabs-website-public-artifacts/isodde_technical_report.pdf.

T. Jain, T. Sun, S. Durand, et al. Biophysical properties of the clinical-stage antibody landscape. *Proceedings of the National Academy of Sciences*, 114(5):944–949, 2017. doi: 10.1073/pnas.1616408114.

T. Jain, T. Boland, and M. Vásquez. Identifying developability risks for clinical progression of antibodies using high-throughput in vitro and in silico approaches. *mAbs*, 15(1):2200540, 2023. doi: 10.1080/19420862.2023.2200540.

H. L. Janssen, J. Hou, T. Asselah, H. L. Chan, F. Zoulim, Y. Tanaka, E. Janczewska, R. G. Nahass, S. Bourgeois, M. Buti, et al. Randomised phase 2 study (jade) of the hbv capsid assembly modulator jnj-56136379 with or without a nucleos (t) ide analogue in patients with chronic hepatitis b infection. *Gut*, 72(7):1385–1398, 2023.

A. Jeddi, M. Ciccone, and B. Taati. LoopFormer: Elastic-depth looped transformers for latent reasoning via shortcut modulation. In *International Conference on Learning Representations*, 2026. URL https://openreview.net/forum?id=RzYXb5YWBs.

D. T. Jones and D. Cozzetto. Disopred3: precise disordered region predictions with annotated protein-binding activity. *Bioinformatics*, 31(6):857–863, 2015.

C. Klein, U. Brinkmann, J. M. Reichert, and R. E. Kontermann. The present and future of bispecific antibodies for cancer therapy. *Nature Reviews Drug Discovery*, 23:301–319, 2024. doi: 10.1038/s41573-024-00896-6.

X. Kong, Z. Zhang, Z. Zhang, R. Jiao, J. Ma, W. Huang, K. Liu, and Y. Liu. Unimomo: Unified generative modeling of 3d molecules for de novo binder design. *arXiv preprint arXiv:2503.19300*, 2025.

M. Konstantinidou and M. R. Arkin. Molecular glues for protein-protein interactions: Progressing toward a new dream. *Cell Chemical Biology*, 31(6):1064–1088, 2024. doi: 10.1016/j.chembiol.2024.04.002.

Z. Kozicka, D. J. Suchyta, V. Focht, G. Kempf, G. Petzold, M. Jentzsch, C. Zou, C. Di Genua, K. A. Donovan, S. Coomar, M. Cigler, C. Mayor-Ruiz, J. L. Schmid-Burgk, D. Häussinger, G. E. Winter, E. S. Fischer, M. Słabicki, D. Gillingham, B. L. Ebert, and N. H. Thomä. Design principles for cyclin K molecular glue degraders. *Nature Chemical Biology*, 20(1):93–102, 2024. doi: 10.1038/s41589-023-01409-z. URL https://doi.org/10.1038/s41589-023-01409-z.

Y. Liao, J. Zhu, J. Xie, L. Lai, and J. Pei. Benchmarking cofolding methods for molecular glue ternary structure prediction. *Journal of Chemical Information and Modeling*, 65(20):11136–11148, 2025. doi: 10.1021/acs.jcim.5c01860.

J. Liu, P. Neupane, and J. Cheng. Improving alphafold2-and alphafold3-based protein complex structure prediction with multicom4 in casp16. *Proteins: Structure, Function, and Bioinformatics*, 94(1):131–141, 2026.

W. Ma, Z. Liu, J. Yang, C. Lu, H. Zhang, and W. Xiao. From dataset curation to unified evaluation: Revisiting structure prediction benchmarks with PXMeter. *bioRxiv*, 2025. doi: 10.1101/2025.07.17.664878. URL https://www.biorxiv.org/content/10.1101/2025.07.17.664878.

A. Maldonado-Contreras, J. R. Birtley, E. Boll, Y. Zhao, K. L. Mumy, J. Toscano, S. Ayehunie, H.-C. Reinecker, L. J. Stern, and B. A. McCormick. Shigella depends on sepa to destabilize the intestinal epithelial integrity via cofilin activation. *Gut Microbes*, 8(6):544–560, 2017.

J. R. Mendell, S. Al-Zaidy, R. Shell, W. D. Arnold, L. R. Rodino-Klapac, T. W. Prior, L. Lowes, L. Alfano, K. Berry, K. Church, et al. Single-dose gene-replacement therapy for spinal muscular atrophy. *New England Journal of Medicine*, 377(18):1713–1722, 2017.

L. S. Mille-Fragoso, J. N. Wang, C. L. Driscoll, H. Dai, T. Widatalla, X. Zhang, B. L. Hie, and X. J. Gao. Efficient generation of epitope-targeted de novo antibodies with germinal. *bioRxiv*, 2025.

M. Mima, C. Mishima-Tsumagari, K. Nakano, M. Morimoto, H. Ogata, M. Sakata, R. Iwaoka, K. Iwata, K. Hachiuma, K. Iwamoto, Y. Fujii, and T. Kurokawa. Structural design of the anti-tnfα therapeutic nanobody® compound, ozoralizumab, to support its potent and sustained clinical efficacy. *Biochemical and Biophysical Research Communications*, 734:150454, 2024. ISSN 0006-291X. doi: https://doi.org/10.1016/j.bbrc.2024.150454. URL https://www.sciencedirect.com/science/article/pii/S0006291X24009902.

J. Ouyang-Zhang, P. Murugan, D. J. Diaz, G. Scarpellini, R. S. Bowen, N. Gruver, A. Klivans, P. Krähenbühl, A. Faust, and M. Al-Shedivat. Triangle multiplication is all you need for biomolecular structure representations. In *The Fourteenth International Conference on Learning Representations*, 2026. URL https://openreview.net/forum?id=CrXcfMLR9q.

M. Pacesa, L. Nickel, C. Schellhaas, et al. One-shot design of functional protein binders with BindCraft. *Nature*, 646:483–492, 2025. doi: 10.1038/s41586-025-09429-6.

S. Passaro, G. Corso, J. Wohlwend, et al. Boltz-2: Towards accurate and efficient binding affinity prediction. *bioRxiv*, 2025. doi: 10.1101/2025.06.14.659707.

X. Peng, R. Guo, F. Guo, Z. Wang, J. Sun, J. Guan, Y. Jia, Y. Xu, Y. Huang, M. Zhang, et al. Unified modeling of 3d molecular generation via atomic interactions with pocketxmol. *Cell*, 189(7): 1904–1922, 2026.

Protenix Team, Y. Zhang, C. Gong, J. Sun, et al. Protenix-v2: Broadening the reach of structure prediction and biomolecular design. *bioRxiv*, 2026a. doi: 10.64898/2026.04.10.717613.

Protenix Team, Y. Zhang, C. Gong, H. Zhang, W. Ma, Z. Liu, X. Chen, J. Guan, L. Wang, Y. Yang, Y. Xia, and W. Xiao. Protenix-v1: Toward high-accuracy open-source biomolecular structure prediction. *bioRxiv*, 2026b. doi: 10.64898/2026.02.05.703733. URL https://www.biorxiv.org/content/early/2026/02/22/2026.02.05.703733.1.

M. Ren, J. Sun, J. Guan, et al. PXDesign: Fast, modular, and accurate de novo design of protein binders. *bioRxiv*, 2025. doi: 10.1101/2025.08.15.670450. Original 2025 report.

Runs N' Poses contributors. Runs N' Poses: Protein–ligand co-folding prediction dataset and benchmark. Official benchmark repository, 2025. URL https://github.com/plinder-org/runs-n-poses. Dataset and benchmark repository.

S. Russell, J. Bennett, J. A. Wellman, D. C. Chung, Z.-F. Yu, A. Tillman, J. Wittes, J. Pappas, O. Elci, S. McCague, et al. Efficacy and safety of voretigene neparvovec (aav2-hrpe65v2) in patients with rpe65-mediated inherited retinal dystrophy: a randomised, controlled, open-label, phase 3 trial. *The Lancet*, 390(10097):849–860, 2017.

N. Saunshi, N. Dikkala, Z. Li, S. Kumar, and S. J. Reddi. Reasoning with latent thoughts: On the power of looped transformers. In *International Conference on Learning Representations*, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/hash/2676109d49d1eb26d6bc584a8f556305-Abstract-Conference.html.

A. Sheikh, T. Wangdi, T. J. Vickers, B. Aaron, M. Palmer, M. J. Miller, S. Kim, C. Herring, R. Simoes, J. A. Crainic, et al. Enterotoxigenic escherichia coli degrades the host muc2 mucin barrier to facilitate critical pathogen-enterocyte interactions in human small intestine. *Infection and immunity*, 90(2):e00572–21, 2022.

B. Shor and D. Schneidman-Duhovny. Combfold: predicting structures of large protein assemblies using a combinatorial assembly algorithm and alphafold2. *Nature methods*, 21(3):477–487, 2024.

K. Singh, M. Burkhardt, S. Nakuchima, R. Herrera, O. Muratova, A. G. Gittis, E. Kelnhofer, K. Reiter, M. Smelkinson, D. Veltri, et al. Structure and function of a malaria transmission blocking vaccine targeting pfs230 and pfs230-pfs48/45 proteins. *Communications biology*, 3(1): 395, 2020.

R. Singh, Y.-H. Kim, S.-J. Lee, H.-S. Eom, and B. K. Choi. 4-1BB immunotherapy: advances and hurdles. *Experimental & Molecular Medicine*, 56:32–39, 2024. doi: 10.1038/s12276-023-01136-4.

K. Stahl, A. Graziadei, T. Dau, O. Brock, and J. Rappsilber. Protein structure prediction with in-cell photo-crosslinking mass spectrometry and deep learning. *Nature biotechnology*, 41(12): 1810–1819, 2023.

K. Stahl, R. Warneke, L. Demann, R. Bremenkamp, B. Hormes, O. Brock, J. Stülke, and J. Rappsilber. Modelling protein complexes with crosslinking mass spectrometry and deep learning. *Nature communications*, 15(1):7866, 2024.

H. Stark, F. Faltings, M. Choi, et al. BoltzGen: Toward universal binder design. *bioRxiv*, 2025. doi: 10.1101/2025.11.20.689494.

W. K. Tang, C. H. Coelho, K. Miura, B. C. N. Tentokam, N. D. Salinas, D. L. Narum, S. A. Healy, I. Sagara, C. A. Long, P. E. Duffy, et al. A human antibody epitope map of pfs230d1 derived from analysis of individuals vaccinated with a malaria transmission-blocking vaccine. *Immunity*, 56(2):433–443, 2023.

A. L. Team and K. Liu. Anewdde: An agentic drug discovery engine for biomolecular interaction modelling and closed-loop design. *bioRxiv*, pages 2026–09, 2026.

B. Team. Geoflow-v2: A unified atomic diffusion model. for protein structure prediction and de novo design. *bioRxiv*, pages 2025–05, 2025.

L. L. Team, H. Kenlay, D. Pretorius, J. Crabbé, A. Bridgland, S. M. Schmon, A. Hilmkil, J. Vuckovic, S. Mathis, T. Matteson, et al. Drug-like antibodies with low immunogenicity in human panels designed with latent-x2. *arXiv preprint arXiv:2512.20263*, 2025.

The OpenFold3 Team. OpenFold3-preview2 technical report. Technical report, 2026. URL https://portal.openfold.omsf.io/reports/of3p2_technical_report.pdf. Second preview release; accessed October 2026.

E. J. Visser, P. Jaishankar, E. Sijbesma, M. A. M. Pennings, E. M. F. Vandenboorn, X. Guillory, R. J. Neitz, J. Morrow, S. Dutta, A. R. Renslo, L. Brunsveld, M. R. Arkin, and C. Ottmann. From

tethered to freestanding stabilizers of 14-3-3 protein-protein interactions through fragment linking. *Angewandte Chemie International Edition*, 62(37):e202308004, 2023. doi: 10.1002/anie.202308004.

W. Wang, Y. Luo, Z. Peng, and J. Yang. Accurate biomolecular structure prediction in casp16 with optimized inputs to state-of-the-art predictors. *Proteins: Structure, Function, and Bioinformatics*, 94(1):142–153, 2026.

A. B. Ward, A. Sali, and I. A. Wilson. Integrative structural biology. *Science*, 339(6122):913–915, 2013. doi: 10.1126/science.1228565.

J. Wohlwend, G. Corso, S. Passaro, et al. Boltz-1: Democratizing biomolecular interaction modeling. *bioRxiv*, 2024. doi: 10.1101/2024.11.19.624167.

K. Wu, T. Schantz, L. Chang, P. Chabria, E. Bashaw, T. R. Kalatturu, D. Y. Kim, A. Chen, N. Corley, A. K. Bera, et al. Design of readers for pathological ptm proteoforms in alzheimer's disease. *bioRxiv*, pages 2026–10, 2026.

S. Xu, Q. Feng, L. Qiao, et al. Benchmarking all-atom biomolecular structure prediction with FoldBench. *Nature Communications*, 17:442, 2026. doi: 10.1038/s41467-025-67127-3.

J. Yan, J. Min-DeBartolo, C.-S. Huang, M. N. Sharif, L. Li, S. Fish, C. Dower, D. Murphy, T. Andreyeva, H. Liu, X. Han, W. Zheng, J. H. Ooi, J. Edmonds, T. Chen, Z. Maben, C. R. Stevens, P. Goihberg, M. Nocula-Lugowska, S. M. Evans, L. Mosyak, K. Kelleher, C. Dickinson, M. Hegen, A. Winkler, and F. Karlsson. Ligand non-competitive GITR antibody prevents formation of the obligatory signal-triggering GITRL: GITR stoichiometry. *Scientific Reports*, 16 (1):2752, 2026. doi: 10.1038/s41598-025-32541-6. URL https://doi.org/10.1038/s41598-025-32541-6.

L. Yang, K. Lee, R. Nowak, and D. Papailiopoulos. Looped transformers are better at learning learning algorithms. In *International Conference on Learning Representations*, 2024. URL https://proceedings.iclr.cc/paper_files/paper/2024/hash/b8402301e7f06bdc97a31bfaa653dc32-Abstract-Conference.html.

V. Zambaldi, D. La, A. E. Chu, et al. De novo design of high-affinity protein binders with AlphaProteo. arXiv:2409.08022, 2024. URL https://arxiv.org/abs/2409.08022.

O. Zhang, X. Zhang, H. Lin, C. Tan, Q. Wang, Y. Mo, Q. Feng, G. Du, Y. Yu, Z. Jin, et al. Odesign: A world model for biomolecular interaction design. *arXiv preprint arXiv:2510.22304*, 2025.

Y. Zhang. I-tasser server for protein 3d structure prediction. *BMC bioinformatics*, 9(1):40, 2008.

Supplementary Material

A. Additional Details on Mol-1

A.1. Architecture, Training and Inference

The exceptional performance of Mol-1 is a product of innovations from data, training as well as inference. We devote substantial effort to data sourcing and curation, systematic quality filtering, and synthetic data generation towards tasks relevant to drug discovery. These efforts are designed to improve data quality and diversity and lead to significant improvement of Mol-1 on tasks across various drug modalities. Regarding experimentally determined structures from the PDB, we only used results released before September 30, 2021 towards fair evaluation.

On the model front, Mol-1 combines increased architectural capacity with greater recycling depth during training to improve learning. We hypothesize that these changes help the model learn underrepresented structural and interaction patterns, which contributes to its performance across diverse benchmarks. We also develop a constraint-conditioned variant that incorporates biological evidence as guidance.

Model Architecture: Scaling Representational Capacity. Mol-1 (577M parameters) integrates diffusion-based structure generation and confidence ranking, conditioned on single and pair representations produced by its trunk. Motivated by recent findings that structure prediction models benefit from increased scale, we expand the pair embedding dimension to 448 channels for diffusion conditioning. Concurrently, to maintain computational efficiency, we design a heterogeneous-width trunk that combines a triangle-attention-free Pairmixer module (similar to Candido et al., 2026; Ouyang-Zhang et al., 2026) with a narrower-width Pairformer architecture. Our ablation studies show that this configuration requires only ~60% of the original FLOPs while achieving comparable fitting performance. Architectural details and fitting performance comparisons are provided in Figure 16.

Training Scheme: Scaling Iterative Computation. Recycling iteratively refines single and pair representations through repeated trunk passes before conditioning structure generation, with additional cycles shown to improve predictions on difficult targets. This mechanism parallels Looped Transformers, where recurrent computation with shared parameters increases effective depth without expanding model size (Geiping et al., 2025; Saunshi et al., 2025). Rather than merely increasing recycling iterations at inference, we progressively shift the training-time cycle distribution toward deeper trajectories—an approach akin to training *Looped Transformers* with scheduled iteration counts (Jeddi et al., 2026; Yang et al., 2024). Concurrently, modern structure prediction models progressively expand training crop-size to capture broader molecular contexts and complex interfaces (Abramson et al., 2024; Passaro et al., 2025; Protenix Team et al., 2026b). We therefore jointly schedule crop size and recycling depth during training: larger crops supply richer structural context, while additional recycling passes provide the extra computation needed to integrate it.

To implement this schedule, we sample the number of recycling iterations at each training stage by discretizing a Beta(α, β) distribution into equal-width bins up to a maximum depth K. Training begins with a cap of $K = 4$ under Beta$(1, 1)$, corresponding to uniform sampling across one to four cycles. As training progresses alongside larger crop sizes, we transition to Beta$(2, 2)$

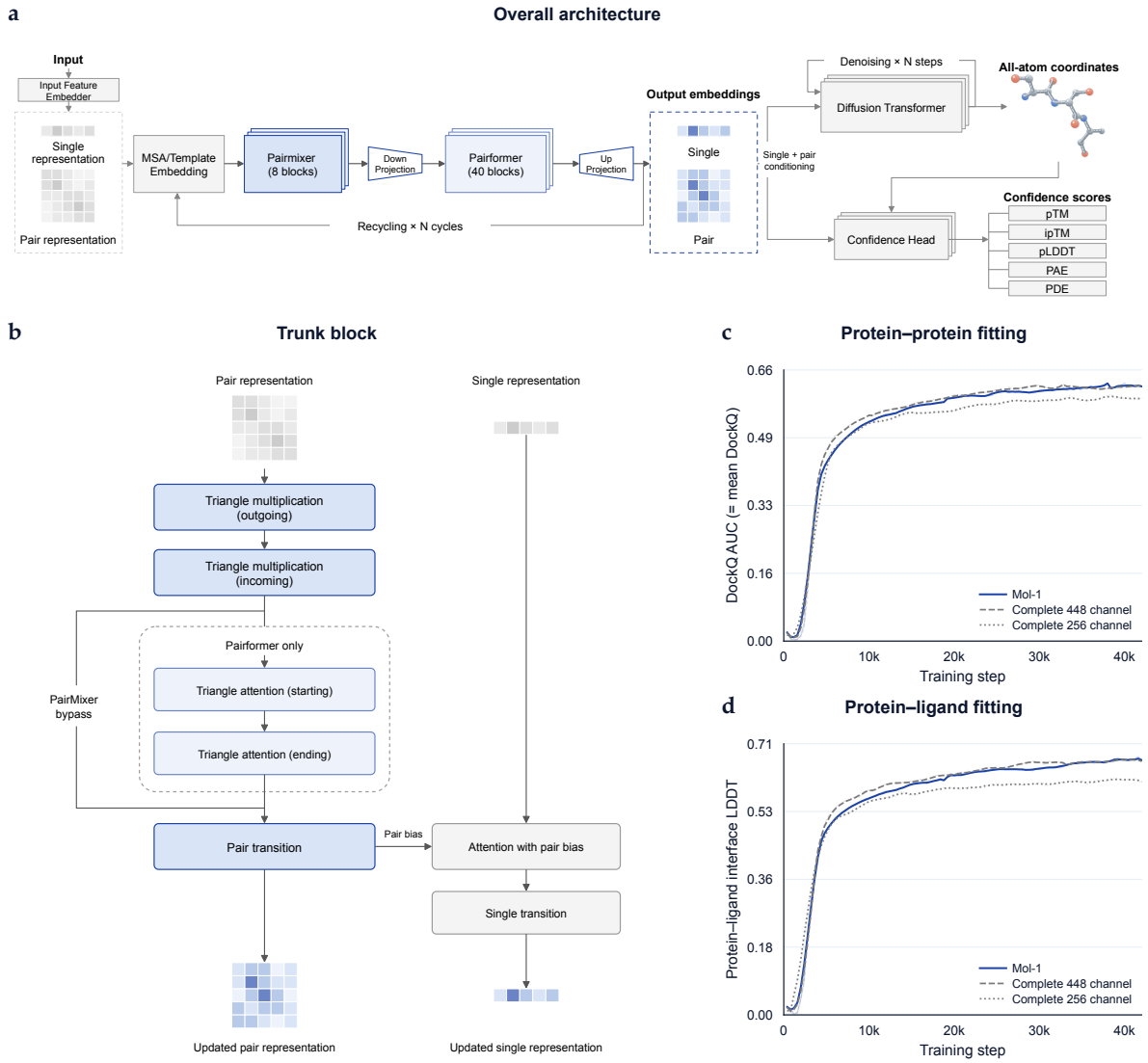


Figure 16 | **Mol-1 architecture and fitting performance.** **(a)** Mol-1 employs a hybrid trunk consisting of 8 Pairmixer-inspired blocks (448 pair channels) followed by 40 full Pairformer blocks (320 pair channels). Learned linear projections bridge the two stages and restore the pair representation to 448 channels at the trunk output. The resulting refined representations condition all-atom structure generation and confidence prediction. **(b)** The Pairmixer-inspired blocks omit triangle attention while retaining outgoing and incoming triangle multiplicative updates, pair transition layers, and single-representation updates. **(c,d)** Training curves comparing Mol-1 against complete 48-block Pairformer baselines (with 448 or 256 pair channels), evaluated by protein–protein DockQ AUC (equivalent to mean DockQ; **c**) and protein–ligand interface lDDT (**d**) on a subset of train data. Mol-1 achieves fitting performance comparable to that of the 448-channel baseline.

to concentrate sampling on intermediate depths (preserving the mean depth), and finally shift to Beta$(2, 1)$ to skew the distribution toward deeper refinement trajectories.

Inference Scaling through Learned Iterative Refinement. By emphasizing deeper refinement trajectories during training, our schedule equips the model to effectively leverage additional computation at inference time. Conventionally, structure prediction models scale inference compute primarily by evaluating more random seeds to broaden candidate sampling (Protenix Team et al., 2026b; The OpenFold3 Team, 2026). However, inference budget can alternatively be allocated to additional recycling passes for deeper iterative refinement. Our results in Figure 1e and 17 demonstrates that the scaling of recycling depth and seed count jointly yields complementary gains, further improving the quality of structure prediction.

A.2. Evaluation details

Curation of MolBench-AB-26 and MolBench-Peptide-26. MolBench-AB-26 is curated following the procedure of curating PXMeter-AB-22to25 in Protenix Team et al. (2026b), but with PDB release date from January 1st, 2026 to September 30, 2026. The complete set contains 242 interfaces in 101 PDB. MolBench-Peptide-26 is curated using a similar workflow to select peptide-containing PDB entries, with two modifications: first, low-homology filtering of polymer-peptide interfaces considers both polymer sequence similarity and peptide sequence identity; second, entries in which the peptide interacts with a crystallographic symmetry mate are excluded. The complete set contains 187 interface clusters in 172 PDB. Evaluation was restricted to the intersection of complexes for which all compared models successfully completed inference, yielding 172 interface clusters from 150 PDB entries.

Antibody–Antigen Benchmark Sources and Comparison Conventions. Mol-1 uses 20 cycles for the four antibody–antigen benchmarks in Figure 1a–d; Mol-1-constraint uses 10 cycles. In Figure 1a–d, AF3 results are taken from FoldBench. Boltz-1, OpenFold3, and Protenix-v1 results on the public benchmarks are from the Protenix-v2 report (Protenix Team et al., 2026a). OpenDDE results are from its own report for FoldBench-AB and PXMeter-AB, and from the Anew report (Team and Liu, 2026) for AF3-AB. FoldBench-AB comprises 104 PDB entries and 160 antibody–antigen interfaces, with metadata obtained from the data released by OpenFold3. Evaluation on FoldBench-AB is conducted using the official FoldBench repository, incorporating a patch to resolve a chain-identification bug.

ESMFold2, Mol-1, Mol-1-constraint, and all MolBench-AB-26 baselines were evaluated in-house. The public-benchmark results do not uniformly use matched cohorts. For example, the PXMeter-AB source cohorts contain 494 entries and 364 clusters for OpenDDE, and 516 entries and 376 clusters for the Protenix-v2-reported baselines, compared with 498 entries and 361 clusters in the supplied Mol-1 evaluation record.

Protein–Ligand and Molecular Glue Evaluation. Success in Figure 3a requires ligand RMSD < 2 Å and lDDT-PLI > 0.8; RNP RMSD is binding-site-superposed and symmetry-corrected. For Figure 3b (MGBench), success requires ligand RMSD < 2 Å and lDDT-PLI > 0.8 and protein-protein interface DockQ > 0.23. Panel c reports individual PoseBusters check pass rates over 64,800 predictions per method. SuCOS-pocket similarity is the percentage product of ligand SuCOS (measuring pose volume and pharmacophore overlap) and binding-pocket residue coverage relative to the closest training-set complex. Higher values indicate greater similarity

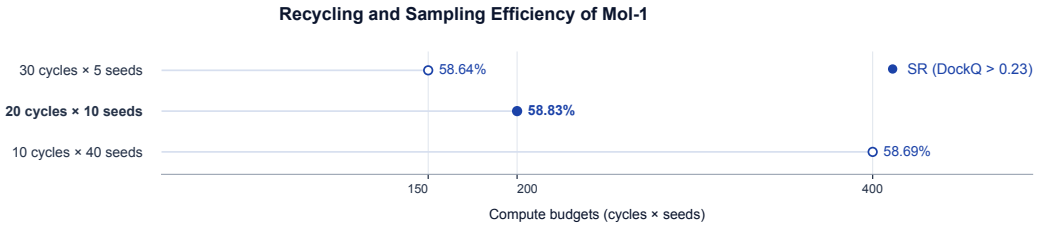


Figure 17 | Recycling and sampling efficiency on MolBench-AB-26. Positions show compute budgets (cycles × seeds); labels give success rates at DockQ > 0.23, with 5 samples per seed.

in ligand pose and pocket coverage. In Figure 3a, parenthesized counts denote the number of systems.

Predictions for Mol-1, Mol-1-constraint, OpenDDE, OpenFold 3, and Protenix-v1 are generated in-house, whereas those for AlphaFold 3 and Boltz-1 are taken from the official RNP release. All models are evaluated under the standard 5 seeds × 5 samples setting (with Mol-1 and Mol-1-constraint using 10 recycling cycles). To ensure a fair comparison, evaluation is restricted to the common intersection of system–ligand pairs across all methods, with top-ranked predictions recomputed using `lig_prot_chain_iptm_average_rmsd`. The constraint model receives additional pocket information.

For MGBench, AF3 and Boltz-1 values are transcribed from the official Figure 3B supplied with the evaluation sources; the other five methods were evaluated in-house. Mol-1 uses 10 cycles. Official sample counts (Low Homology/High Homology/Seen) are 23/59/109 for AF3 and 25/60/112 for Boltz-1, versus 21/59/102 in the internal record. These comparisons do not uniformly use matched cohorts.

PoseBusters results use the sample-level intersection of Mol-1, OpenDDE, and OpenFold3 on RNP: 64,800 predictions per method, with five seeds and five samples per seed. We follow the validity definition in the Protenix-v2 report, including its newly introduced checks for sp2-center flatness, amide flatness, and sp3-center non-flatness. All checks are shown except RMSD and the three input-loading checks, which pass for all samples.

Figure 4 shows only the target protein pair and the specified molecular glue. DockQ evaluates the protein–protein interface; ligand RMSD measures ligand placement after binding-pocket alignment, using the pocket definitions from OpenStructure (Biasini et al., 2013).

A.3. Additional Study of Antibody-Antigen Structure Prediction Performance

Antibody Inference-Time Sampling and Seed-Selection Variability. For Figure 1f, we perform sensitivity analysis to seed selection for Mol-1 on FoldBench-AB. For Figure 1e, each pair is predicted with K seeds and five samples per seed; the prediction with the highest ipTM-based confidence score is selected from the resulting $K \times 5$ candidates. The horizontal axis is logarithmic. Shaded bands show the 5th–95th percentiles over seed subsets, using 256 draws without replacement for intermediate K. At $K = 1$, all seeds are enumerated; $K = 100$ uses the full pool and has no percentile band. The curve labeled Mol-1 uses 20 cycles; the 10- and 30-cycle variants are labeled separately. Figure 1f enumerates all $\binom{32}{5} = 201{,}376$ five-seed combinations on a fixed set of 160 FoldBench-AB interfaces. Success rate is the number of successful interfaces divided by 160; bar heights and labels give the number of combinations. Dark blue bars identify combinations achieving at least the reported 75% success rate: 125,300 combinations, or 62.22%

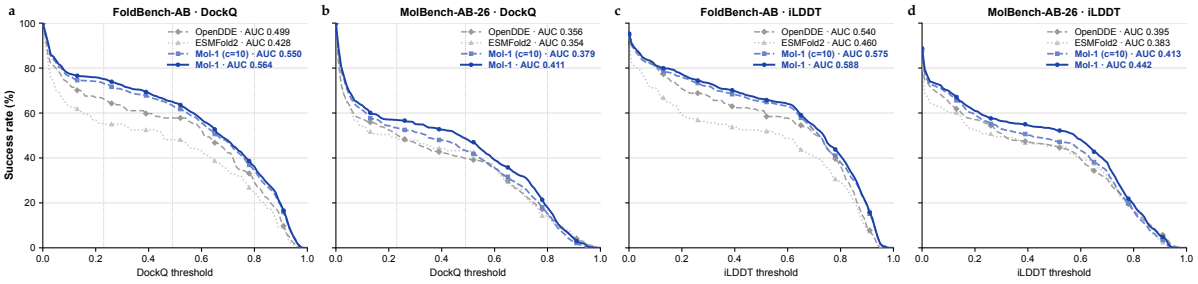


Figure 18 | Antibody–antigen prediction on FoldBench-AB and MolBench-AB-26 with 5 seeds. **a–d,** Success-rate curves for DockQ (**a,b**) and interface LDDT (**c,d**) on FoldBench-AB (**a,c**) and MolBench-AB-26 (**b,d**). Each curve gives the fraction of interfaces (**a,c**) or interface clusters (**b,d**) whose selected prediction attains the metric above threshold $t \in [0, 1]$. All methods are ranked by ipTM-based confidence ranking. Mol-1 uses 10 or 20 cycles as labeled and is shown in blue; baselines are shown in gray. Vertical dotted lines in **a,b** mark the DockQ thresholds 0.23, 0.49, and 0.80; the area under each curve (AUC) is reported in the legend.

of the total. This distribution measures seed-selection variability on the fixed test set, rather than uncertainty from resampling test interfaces.

Fine-grained Comparisons of DockQ and iLDDT Metrics. Fig. 18 compares success-rate profiles against OpenDDE (antibody–antigen finetuned checkpoint with the official no-template configuration) and ESMFold2 (official configuration with 20 cycles) on FoldBench-AB and MolBench-AB-26. The curves for Mol-1 lie above both baselines at nearly all DockQ thresholds, and the ordering of the areas under the curves (AUCs) is consistent across benchmarks and metrics: on FoldBench-AB DockQ, Mol-1 reaches 0.564 versus 0.499 for OpenDDE and 0.428 for ESMFold2; on MolBench-AB-26 DockQ, 0.411 versus 0.356 and 0.354. The same pattern holds for interface LDDT (0.588 versus 0.540 and 0.460 on FoldBench-AB; 0.442 versus 0.395 and 0.383 on MolBench-AB-26). Reducing the latent budget from 20 to 10 cycles costs 0.01–0.03 AUC points, indicating that cycle count acts as a monotonic, transferable scaling knob rather than a benchmark-specific tuning choice. Comparing the 20- and 10-cycle configurations, the separation is largest in the intermediate regime around DockQ $\approx$ 0.35–0.50, which marks the transition from acceptable to medium-quality poses, and the gap narrows above 0.8, suggesting that additional recycling primarily refines coarse docks into medium-accuracy poses rather than shifting top-tier atomic accuracy.

Test-Time Scaling. Fig. 19a compares how prediction accuracy scales with the number of inference seeds on MolBench-AB-26. Mean DockQ rises monotonically with the number of seeds K for every method, but with markedly different efficiency. Mol-1 gains from 0.359 at $K = 1$ to 0.463 at $K = 100$ (+0.104), and the 30-cycle variant from 0.368 to 0.469; OpenDDE and ESMFold2 plateau at 0.384 and 0.390, reaching within 0.01 of their $K = 100$ values by $K \approx 20$ and $K \approx 40$, respectively. The practical consequence is that Mol-1 surpasses OpenDDE and ESMFold2's full **100-seed** pool with as few as **2 seeds**.

Fig. 20 compares 10-, 20-, and 30-cycle configurations under the identical 5-seed protocol. Doubling from 10 to 20 cycles lifts the DockQ $\geq$ 0.23 success rate by +1.6 points on FoldBench-AB (73.4% $\rightarrow$ 75.0%) and by +3.8 points on MolBench-AB-26 (53.5% $\rightarrow$ 57.3%), with parallel gains at the 0.49 and 0.80 strata, so the deeper representation refinement improves high-accuracy poses

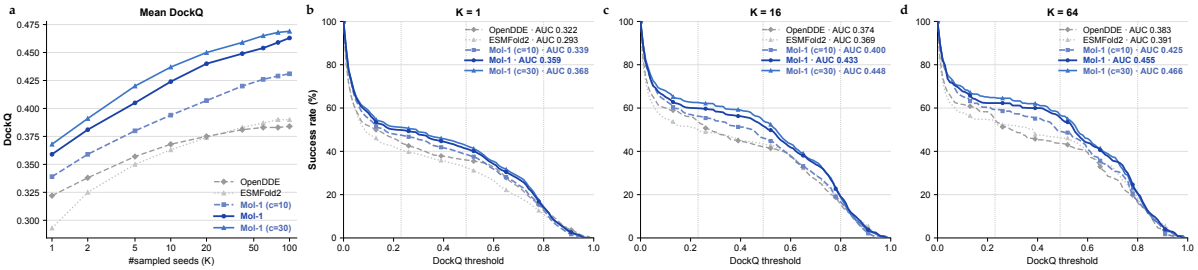


Figure 19 | Antibody–antigen test-time scaling on MolBench-AB-26. **a,** Mean DockQ versus the number of sampled seeds K with a logarithmic seed-count axis. For each pair, the candidate with the highest confidence is selected from K seeds × 5 samples; per-pair values are averaged within each interface cluster and then across clusters. Mol-1 uses 10, 20, or 30 cycles as labeled and is shown in blue; baselines are shown in cool gray. Intermediate K values use 256 draws without replacement; $K = 1$ enumerates all seeds, and $K = 100$ is the exact full-pool result. **b–d,** DockQ success-rate curves at $K = 1$, 16, and 64 sampled seeds, using the same per-pair selection and cluster-level aggregation as in **a**: each curve gives the fraction of clusters whose selected prediction attains DockQ > t across thresholds $t \in [0, 1]$. Mol-1 with 30 cycles attains the highest success rate at every seed budget, followed by 20 and 10 cycles; all Mol-1 variants outperform OpenDDE and ESMFold2. ESMFold2, OpenDDE, and all Mol-1 variants were evaluated in-house.

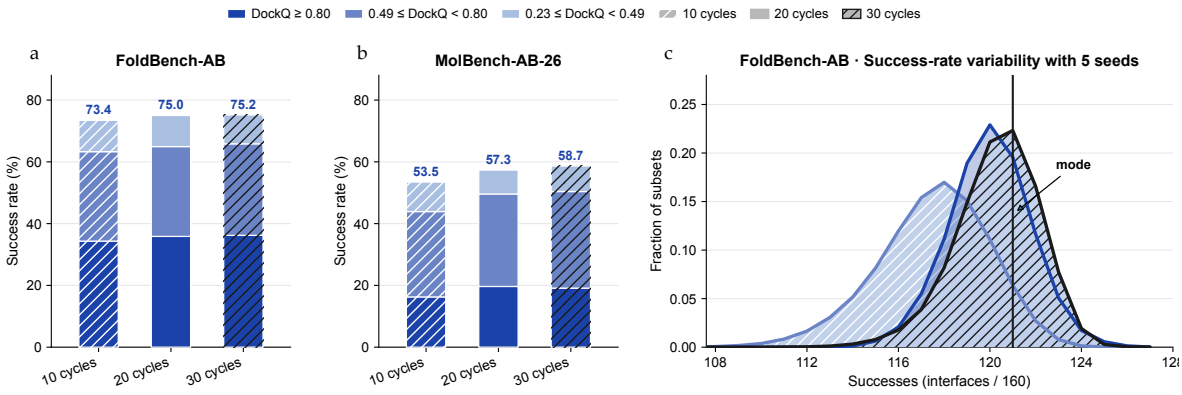


Figure 20 | **a, b,** Effect of cycles (10, 20, 30) on Mol-1 success rates with 5 seeds for **a**, FoldBench-AB and **b**, MolBench-AB-26. Bars are stacked by DockQ stratum, from dark to light: DockQ ≥ 0.80, 0.49 ≤ DockQ < 0.80 and 0.23 < DockQ < 0.49 threshold bands, and textured by cycle count (white hatching, 10 cycles; solid fill, 20 cycles; black hatching, 30 cycles; 20 cycles is the default Mol-1 setting). Numbers above bars denote the overall DockQ > 0.23 success rate. Values are exact means over all $\binom{32}{5}$ five-seed subsets of the 32-seed pool, with the best sample per interface chosen by ipTM-based confidence ranking. **c**, Success-rate variability with 5 seeds on FoldBench-AB: distribution of the DockQ > 0.23 success rate across the same $\binom{32}{5} = 201{,}376$ seed subsets (kernel-smoothed); the arrow marks the mode of the 30-cycle distribution (75.625% success rate).

rather than only marginal ones. On MolBench-AB-26, where sampling is further from saturation, cycle scaling is nearly as effective as adding seeds, making it the more compute-efficient lever at small budgets. Extending to 30 cycles yields marginal gains, so we fix 20 cycles as the default.

B. Mol-1-constraint: Enhanced Prediction with Biological Evidence

Biological experiments can provide partial structural information about biomolecular interactions. For example, NMR chemical shift perturbations or mutations can identify candidate binding-site residues, yielding residue-level interface constraints (Dominguez et al., 2003), while cross-linking mass spectrometry (Stahl et al., 2023, 2024) can identify proximal residue pairs, yielding distance constraints. We developed Mol-1-constraint to incorporate such evidence into structure prediction through learned constraint conditioners.

B.1. Constraint-Aware Training Regime

Mol-1-constraint is finetuned from Mol-1 using structure-derived constraints across protein–protein, protein–ligand, protein–DNA, and protein–RNA complexes. The whole training combines three complementary constraint types: binding site (SITE) specifies binding-site residues that should contact a designated partner; forbidden site (EXCLUSION) specifies residues that should remain separated from the partner; and contact (CONTACT) specifies token pairs that should be in contact. During training, constraints are randomly sampled online from reference structures. SITE conditioning alternates between the full set of available interface sites and an empty input with equal probability, while EXCLUSION and CONTACT use separate sampling and dropout controls. This exposes the model to individual constraints, their combinations, and empty constraint inputs within the same training regime.

The constraint-conditioning modules are trained jointly with partial Mol-1 backbone blocks, while the remaining pretrained parameters are frozen. Constraint conditioning modules use zero-initialized output projections, so their initial contribution to the pretrained representations is zero. The training objective retains the native prediction losses and adds constraint-dependent auxiliary penalties:

$$\mathcal{L} = \mathcal{L}_{\text{native}} + 0.6\,\mathcal{L}_{\text{site}} + 0.1\,\mathcal{L}_{\text{excl}} + 0.1\,\mathcal{L}_{\text{contact}}. \tag{1}$$

The SITE term penalizes supplied residues whose predicted minimum heavy-atom distance to the partner exceeds 4.5 Å; the EXCLUSION term penalizes supplied residues whose corresponding distance falls below 6.0 Å; and the CONTACT term penalizes supplied token pairs whose minimum heavy-atom distance exceeds 4.5 Å. Each auxiliary term is evaluated only on its supplied, eligible constraints and is zero when that set is empty. These hinge penalties encourage satisfaction of the requested geometry without requiring an exact reference distance within the permitted region.

Constraint Auxiliary Objectives. Let $\hat{\mathbf{x}}^{(m)}$ denote the predicted coordinates for diffusion training sample $m \in \{1 \dots, M\}$. For a source residue r and its designated partner P, define

$$d_{rP}^{(m)} = \min_{a \in H_r,\, b \in H_P} \left\| \hat{\mathbf{x}}_a^{(m)} - \hat{\mathbf{x}}_b^{(m)} \right\|_2, \tag{2}$$

where H_r and H_P are the applicable heavy-atom sets. Similarly, $d_{ij}^{(m)}$ is the minimum heavy-atom distance between tokens i and j. Writing $[z]_+ = \max(z, 0)$, the auxiliary losses are

$$\mathcal{L}_{\text{site}} = \frac{1}{M|\mathcal{S}|} \sum_{m=1}^{M} \sum_{r \in \mathcal{S}} \frac{[d_{rP}^{(m)} - 4.5]_+}{4.5}, \tag{3}$$

$$\mathcal{L}_{\text{excl}} = \frac{1}{M|\mathcal{E}|} \sum_{m=1}^{M} \sum_{r \in \mathcal{E}} \frac{[6.0 - d_{rP}^{(m)}]_+}{6.0}, \tag{4}$$

$$\mathcal{L}_{\text{contact}} = \frac{1}{M|C|} \sum_{m=1}^{M} \sum_{(i,j) \in C} [d_{ij}^{(m)} - 4.5]_+. \tag{5}$$

All distances are expressed in angstroms. Here, $\mathcal{S}$ and C contain the supplied constraints that remain supervisable under the reference coordinate mask. Their distances use predicted coordinates restricted to resolved heavy atoms. In contrast, EXCLUSION uses all predicted heavy atoms of the supplied residues and partner, without applying the reference coordinate mask. Empty supervised sets contribute zero. SITE and EXCLUSION violations are normalized by their distance thresholds; CONTACT violations are not normalized.

Native Structural Losses. The retained structural training objective (aside from the constraint auxiliary objectives above) is

$$\mathcal{L}_{\text{native}} = 4.0\,(\mathcal{L}_{\text{MSE}} + \mathcal{L}_{\text{smooth-lDDT}}) + 0.03\,\mathcal{L}_{\text{distogram}}. \tag{6}$$

The confidence head is kept frozen, and its inputs are detached from the backbone (stop-gradient); confidence losses therefore do not contribute gradients to the trainable parameters.

B.2. Training Data and Reference Constraint Sampling

Training Data Mixture. Training uses the configured training subsets among biomolecular complexes, with sampling probabilities over different interfaces types:

Table 1 | Data sampling ratio for Mol-1-constraint training.

Interface type	Sampling probability
Protein-protein	0.50
Protein-ligand	0.40
Protein-DNA	0.05
Protein-RNA	0.05

Sampling of SITE. For protein–protein interfaces (including Ab-Ag and protein-peptide), the source side is randomly sampled online; for protein–nonprotein interfaces, the protein always provides the source residues. The sampling probability is shown in Table 1. SITE annotations are constructed from resolved heavy-atom contacts within 4.5 Å in the full assembly and subsequently projected into the training crop of 384. With probability 0.5, all available oracle sites in the crop are supplied; otherwise, the SITE input is empty. Intermediate coverage levels, false-positive augmentation, and bilateral SITE sampling are disabled. Thus, "full" coverage refers to the available sites after crop projection, rather than every interface residue in the uncropped assembly.

Sampling of EXCLUSION. EXCLUSION is sampled **alongside SITE** throughout training, with an independent dropout probability of $p = 0.5$ (i.e., no EXCLUSION constraint applied half of the time). Eligible source residues lie on the surface, outside the full SITE oracle, and have no annotated unresolved heavy atoms. Before cropping, their minimum resolved heavy-atom distance must exceed 6.0 Å both to the designated partner and to the source oracle residues. After crop projection, the sampler requests a connected patch of 3–12 residues, using an 8.0 Å residue-centroid adjacency threshold. A smaller available connected component may yield fewer than the requested number of residues; if no component contains at least three eligible residues, the EXCLUSION input is kept empty.

Sampling of CONTACT. CONTACT uses binary token-pair constraints independent from SITE/EXCLUSION with a fixed heavy-atom 4.5 Å threshold. All CONTACT conditions are independently dropped out with probability $p = 0.5$; otherwise, the requested number of pairs is sampled uniformly from $K = \{1, 2, 3, 4\}$. Candidate pairs are derived from resolved heavy-atom contacts after cropping, within the selected source and partner chains. During training, both inter-chain and intra-chain contact candidates are considered and they have equal sampling weights, re-normalized over nonempty pools, and pairs are sampled without replacement. Same-chain contacting residue pairs must have an original sequence separation greater than 5 residues. If fewer eligible pairs exist, the contact sampler returns fewer constraints.

B.3. Model Architecture and Optimization.

Model Architecture. We incorporate SITE, EXCLUSION, and CONTACT constraints through lightweight conditioners that produce additive updates to the initial pair representation, $z_{\text{init}} \in \mathbb{R}^{N \times N \times c_z}$. SITE and EXCLUSION use the same encoder architecture with independent modules: outer products of the partner mask (over the whole chain) and the source residue-level mask are constructed, each associated with a learned c_z-dimensional embedding, followed by a bias-free linear projection. CONTACT instead encodes explicitly specified contacting token pairs using a learned embedding and a bias-free linear projection, scattering the resulting features symmetrically to entries (i, j) and (j, i). The enabled constraint embeddings are summed and added to z_{init} before processing by the backbone. All output projections are **zero-initialized**, preserving the initial backbone representation, and mask gating ensures that empty constraints produce exactly zero updates throughout training.

Parameter Optimization. The run starts with Mol-1 checkpoint. The trainable parameters comprise the constraint conditioner and last 8 Pairformer blocks, totaling 48.7M parameters out of 577.4M. Training is distributed across 128 GPUs using distributed data parallelism. We use Adam with $(\beta_1, \beta_2) = (0.9, 0.95)$, weight decay 10^{-8}, and gradient-norm clipping at 10. Peak learning rates are 6×10^{-3} uniformly for the constraint conditioners and 3×10^{-4} for the unfrozen pretrained Pairformer blocks, with both groups updated from the start of fine-tuning. A shared schedule applies linear warmup followed by cosine decay. For zero-based scheduler index t, the group learning rate is $\eta_g(t) = \eta_g^{\max} f(t)$, where

$$f(t) = \begin{cases} (t+1)/101, & 0 \le t \le 100, \\ 0.1 + 0.45 \left[1 + \cos\left(\pi \frac{t-100}{1900}\right)\right], & 100 < t \le 2000. \end{cases} \tag{7}$$

Training uses 384-token crops, 32 diffusion training samples per structure, bfloat16 precision, and one microbatch per GPU per optimizer update, without gradient accumulation. The final model uses the EMA 0.999 checkpoint saved after 2,000 iterations.

B.4. Binding-site Annotations for Benchmark Results

Constraint annotations were derived from reference PDB structures. At present, we only evaluate the binding-site constraint as input. On the *source* chain, a protein residue was annotated if any of its resolved heavy atoms lay within 4.5 Å of a resolved heavy atom on the designated *partner*. During inference, constraints were supplied as a residue-level mask on only the source side of each interface. Formally, the constraint input was represented as $(\text{source_chain_id}, \text{partner_chain_id}, C_{\text{source}})$, where $C_{\text{source}} = \{i_1, i_2, \ldots, i_k\}$ denotes the set of annotated residue indices identifying potential binding sites on the source chain.

For **antibody–antigen complexes**, we used FoldBench-AB, PXMeter-AB, MolBench-AB-26 and AF3AB. The antigen chain served as the source chain, and antigen residues contacting the antibody interface were annotated as epitope residues. Antibody and antigen roles were assigned using SAbDab annotations; For **protein–ligand complexes**, we used FoldBench-PL and Runs N′ Pose. The protein receptor served as the source chain, and residues contacting the designated ligand were annotated as binding-pocket residues. For **protein–peptide complexes**, we used MolBench-Peptide-26. The receptor protein served as the source chain, and residues contacting the peptide were annotated as binding-site residues. Dataset-defined roles were retained where available; otherwise, the longer chain was assigned as the receptor and the shorter as the peptide. No equal-length pairs were encountered.

B.5. Accuracy and Robustness of Constraint-Guided Prediction

Main text has shown the strong performance of Mol-1-constraint on several evaluation set (eg., Fig. 1 and 3). To further justify the effectiveness of Mol-1-constraint, we evaluate it on FoldBench-AB, alongside prior constraint-enabled models, Protenix v0.5 (ByteDance AML AI4Science Team et al., 2025), Chai-1 (Chai Discovery Team, 2024), Boltz-1 (Wohlwend et al., 2024), and Boltz-2 (Passaro et al., 2025), all under constraint-conditioned settings for fair comparison. For each target and binding-site condition, each model generated five diffusion samples for each of five random seeds. The candidate with the highest pair ipTM confidence score among the 25 samples was selected to represent each target for evaluation. As shown in Fig. 21 (a), Mol-1-constraint achieved success rates of 88.3% at DockQ $>$ 0.23 and 45.4% at DockQ $>$ 0.80 across all 160 shared interfaces with reference binding-site as conditions. Boltz-2, the second-best baseline model at both thresholds, achieved 71.9% and 25.6%, respectively. These results correspond to improvement margin of 16.4 and 19.8 percentage points, indicating that Mol-1-constraint exhibits state-of-the-art prediction accuracy for constraint-conditioned antibody-antigen prediction. To assess the robustness to incomplete and incorrect site constraint, we varied both the coverage of *correct* site annotations and the amount of added *incorrect* site annotations.

Conditional Site Construction. For each interface, we defined the reference interface residue set (*correct*) $\mathcal{T}$ on the annotated source side using a minimum heavy-atom distance of 4.5 Å to the designated partner in the reference (Ground-truth, GT) structure. The *incorrect*-site pool $\mathcal{F}$ comprised resolved source-side residues outside $\mathcal{T}$ whose minimum distance to the partner at least exceeded 4.5 Å. For each correct-site coverage c, we sampled $k = \lceil c|\mathcal{T}| \rceil (c \in \{0, 0.1, 0.3, 0.6, 1.0\})$ correct residues uniformly without replacement. To introduce incorrect site, we retained the sampled correct residues and added $m = \lceil rk \rceil$ residues sampled uniformly without replacement from $\mathcal{F}$, where $r \in \{0.25, 0.5, 1.0\}$. These additions were evaluated at full $c = 1.0$ and partial $c = 0.6$. The corresponding conditions without incorrect site served as the $r = 0$ references.

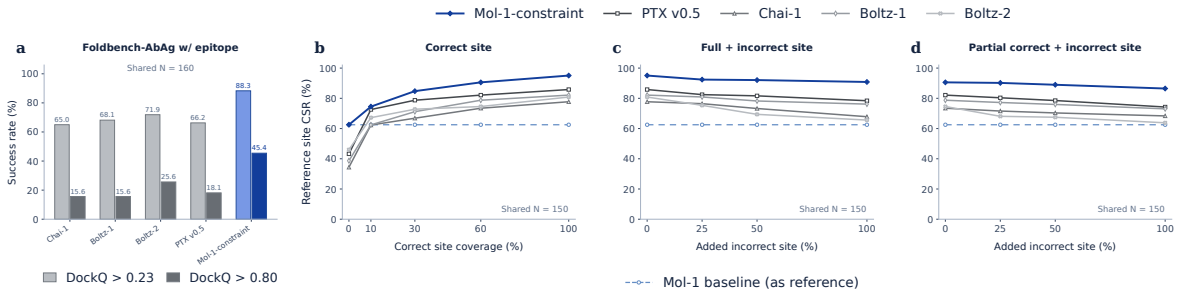


Figure 21 | **Accuracy and robustness of constraint-guided antibody–antigen interface prediction. (a)** Success rates at DockQ > 0.23 and DockQ > 0.80 with full correct site on 160 shared FoldBench interfaces. The reported value is the mean success rate of all five seed combinations in a 32-seeds pool. **(b)** Mean reference site CSR across correct-site coverage levels. **(c, d)** Mean reference site CSR with increasing incorrect site added to full=100% (**c**) or partial=60% (**d**) correct-site coverage. Panels **(b)–(d)** use the same 150 interfaces. The dashed line denotes the Mol-1 reference without constraints.

Constraint Satisfaction Evaluation Metrics. We measured predictions using the reference site constraint satisfaction rate (**CSR**), defined as the fraction of the GT reference interface residue set that contacts the designated partner in the predicted structure under the same 4.5 Å criterion. This metric measures the **recovery** of full reference residue set under different levels of corruption constraint condition as input. For the selected prediction, reference site CSR was computed as

$$\mathrm{CSR} = \frac{1}{|\mathcal{T}|} \sum_{i \in \mathcal{T}} \mathbf{1}\left[\min_{\substack{a \in H(i) \\ b \in H(P)}} \|\hat{\mathbf{x}}_a - \hat{\mathbf{x}}_b\|_2 \leq 4.5 \text{ Å} \right], \tag{8}$$

where $\mathbf{1}$ is an indicator function, $H(i)$ and $H(P)$ denote the heavy atoms of source residue i and the designated partner **on the GT interface**, respectively, and $\hat{\mathbf{x}}$ denotes predicted coordinates. Reported values are means of per-interface reference site CSR. Each model generated 25 candidates (5 seeds × 5 samples) per target and condition, the candidate with the highest paired ipTM score was selected.

We report the *mean* per-interface reference site CSR on a common subset of 150 interfaces across all models and constraint conditions. The remaining 10 interfaces were excluded because the corruption protocol was inapplicable across all models: 8 had insufficient non-interface residues to meet the prescribed false-positive quota (IDs: 7UCX A–B, 8CHE A–E, 8DUZ A–F, 8FBW A–C, 8FDD A–C, 8GHO A–C, 8T59 A–E, and 8TMZ B–C), one had no non-interface residues (ID: 8FAS A–C), and one had an empty reference site set (ID: 8ONI A–E).

Mol-1-constraint's mean CSR increased from 65.5% without site constraint to 75.1% at 10% correct-site coverage and 94.9% at full coverage (Fig. 21 b). It outperformed all compared methods at every tested nonzero coverage level. Much of this improvement was retained when incorrect site annotations were added. As added incorrect site increased from 0% to 100%, CSR declined from 94.9% to 90.5% with full correct site (Fig. 21 c), and from 90.3% to 85.8% with 60% correct-site coverage (Fig. 21 d). Mol-1-constraint maintained the highest CSR throughout both analyses, supporting robustness to incomplete interface annotations and randomly added non-interface annotations under the tested conditions.

B.6. Epitope-constrained prediction of antigen–antibody complexes

In these exploratory case studies, we assessed whether sparse epitope hypotheses informed by published literature evidence and structure-related reasoning could support accurate prediction with Mol-1-constraint. Candidate residues were finally chosen through qualitative assessment of prior biological knowledge, solvent accessibility, and monomeric model confidence. We compared predictions conditioned on these hypotheses with Mol-1 (unconstrained) and predictions conditioned on reference-derived *oracle* epitopes. Three test antibody–antigen complexes are considered (all released after 2021-09): 7UA2 (FoldBench-AB), 9Z76 and 9GEI (MolBench-AB-26).

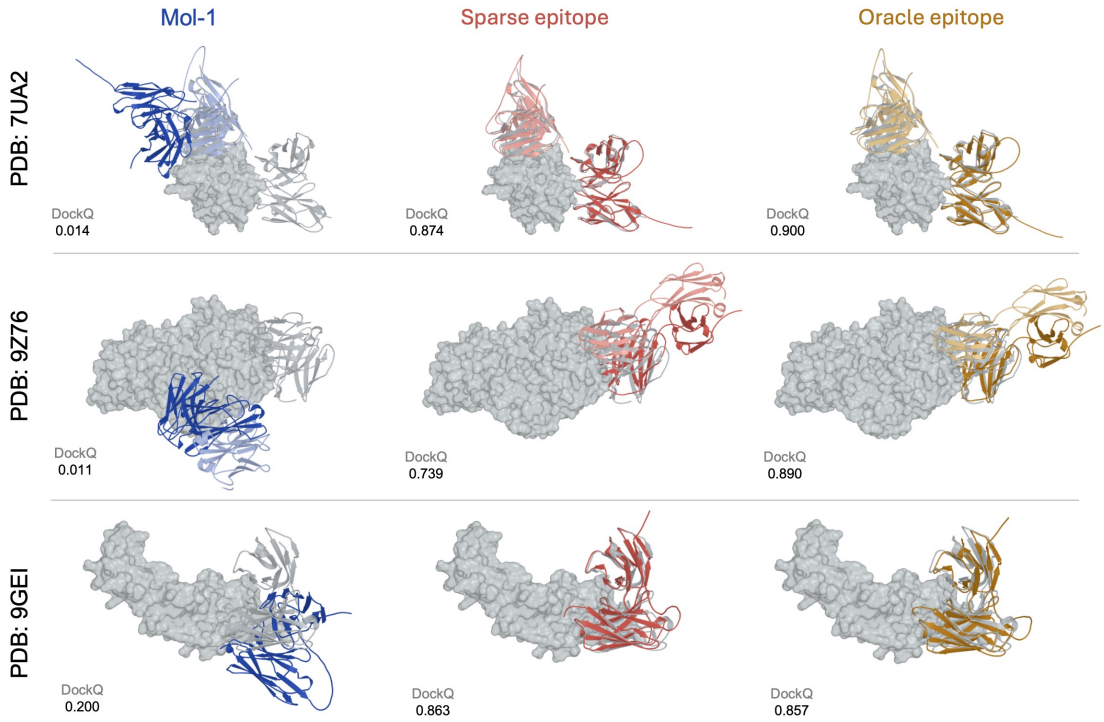


Figure 22 | Mol-1-constraint with literature-guided sparse epitope constraints improves Mol-1 predictions for antigen–antibody complexes. Rows show Pfs230D1 bound to LMIV230-01/230AL-18 (PDB 7UA2; a), EatA bound to antibody 25 Fab (PDB 9Z76; b), and a Gal/GalNAc lectin fragment bound to CP33-H/L-LA22 scFv (PDB 9GEI; c). Columns compare Mol-1 (blue), Mol-1-constraint with sparse binding sites (only three residues, coral), and reference-derived epitope (4.5 Å cutoff, ochre). Reference antigens are colored in gray; predicted antibodies are shown as superposed cartoons. Each displayed prediction was selected by chain-pair ipTM top-1 among 25 candidates.

PDB 7UA2. This complex contains the malaria transmission-blocking vaccine antigen Pfs230D1 bound to two distinct single-chain variable fragments (scFvs), LMIV230-01 and 230AL-18 (Tang et al., 2023). For 230AL-18, the reported epitope overlap with the 4F12 Fab footprint (PDB 6OHG) (Singh et al., 2020) suggested a candidate β-strand/loop region on Pfs230D1. Combining this regional evidence with structural assessment of the monomeric antigen, we proposed D580/Y582/F595 as a sparse epitope hypothesis (PDB `auth_seq_id` numbering throughout). For the Pfs230D1–230AL-18 interface, the reported prediction achieved a DockQ of 0.874, compared with 0.014 for Mol-1 and 0.900 with the oracle epitope.

PDB 9Z76. This complex contains EatA, an enterotoxigenic *Escherichia coli* mucinase, bound

to monoclonal antibody 25 Fab (Buckley et al., 2026). Structural studies of the SPATE-family homolog SepA support a β-helical passenger-domain scaffold with inserted serine protease and domain-of-unknown-function (DUF) subdomains (Maldonado-Contreras et al., 2017). Prior studies implicate the DUF in MUC2 binding and efficient mucin degradation (Sheikh et al., 2022) and identify the serine protease domain as an antibody target (Buckley et al., 2025, 2026). These findings provided structural and functional context but did not establish the location of the antibody 25 epitope. Within this broader context, structural assessment of the monomer led us to propose Q10/N35/G623, a sequence-dispersed but spatially clustered triplet on the β-helical scaffold, as an exploratory epitope hypothesis. Conditioning on this sparse set yielded a reported prediction with a DockQ of 0.739, compared with 0.011 for Mol-1 and 0.890 with the oracle epitope.

PDB 9GEI. This complex contains a fragment of the *Entamoeba histolytica* Gal/GalNAc lectin, which participates in host-cell adhesion, bound to the CP33-H/L-LA22 scFv (Gérard et al., 2026). Prior mapping localized the epitopes of two adherence-enhancing and one complement-sensitizing monoclonal antibodies to residues 882–959 of the lectin heavy subunit (Dodson et al., 1997). Although this evidence does not establish the epitope of CP33-H/L-LA22, it provided a rationale for exploring this segment as a candidate binding region. Combining this evidence with structural assessment of the antigen monomer, we proposed V938/K946/K953 within this segment as a sparse epitope hypothesis. The reported prediction achieved a DockQ of 0.863, compared with 0.200 for unconstrained Mol-1 and 0.857 with the oracle epitope, indicating comparable interface accuracy for the reported sparse- and oracle-conditioned predictions.

Each prediction run contained 25 candidates per complex/model, corresponding to 5 random seeds and 5 diffusion samples per seed. Within each condition, we selected the candidate with the highest ipTM for the target chain pair: A–C for 7UA2 and 9Z76, and A–B for 9GEI. DockQ was used to evaluate the selected prediction, as shown in Fig. 22. Table 2 additionally reports DockQ summaries for each model including OpenDDE and ESMFold2: the mean, the ipTM-ranked, and the oracle DockQ among the 25 candidates. Target interfaces are A–C for 7UA2 and 9Z76, and A–B for 9GEI.

C. VHH Design and Experimental Methods

C.1. Recombinant Protein Construction, Expression, and Purification

Each VHH candidate was fused to the N-terminus of a human IgG1 Fc domain containing the C220S substitution via a G4S linker. Control monoclonal antibodies were expressed as full-length IgG molecules. Coding sequences were synthesized and cloned into pcDNA3.4. Proteins for subsequent characterization were transiently expressed in CHO or Expi293pro cells in serum-free medium at 37 °C and 8% CO_2. Expression cultures ranged from 1 to 4 mL, typically yielding several hundred micrograms to a few milligrams of protein per culture. Supernatants were harvested 5–7 days after transfection, clarified by centrifugation, and filtered through a 0.22 μm membrane.

VHH-Fc fusions and control antibodies were purified by Protein A affinity chromatography. After washing with PBS (pH 7.4), bound proteins were eluted with 10 mM glycine (pH 1.5), immediately neutralized with 1 M Tris, and buffer-exchanged into PBS. Purity was evaluated by non-reducing SDS-PAGE, and monomer content and aggregation were assessed by size-exclusion high-performance liquid chromatography (SEC-HPLC). Purified batches with SEC-HPLC purity > 90% were used for subsequent characterization.

Table 2 | Constraint case-study results with five seeds and five samples per seed. Mean averages all 25 candidates; ipTM top-1 reports the DockQ of the candidate with the highest target chain-pair ipTM; Best-of-25 reports the maximum DockQ within the same pool. Mol-1C abbreviates the Mol-1-constraint model with different input conditions.

Target interface	Model / condition	Mean	ipTM top-1	Best-of-25
7UA2 A–C	OpenDDE	0.046	0.046	0.153
	ESMFold2	0.080	0.055	0.411
	Mol-1	0.014	0.014	0.014
	Mol-1C Sparse Epitope	0.446	0.874	0.891
	Mol-1C Oracle Epitope	**0.624**	**0.900**	**0.905**
9Z76 A–C	OpenDDE	0.011	0.004	0.059
	ESMFold2	0.021	0.009	0.068
	Mol-1	0.012	0.011	0.039
	Mol-1C Sparse Epitope	0.623	0.739	0.847
	Mol-1C Oracle Epitope	**0.871**	**0.890**	**0.912**
9GEI A–B	OpenDDE	0.041	0.026	0.239
	ESMFold2	0.012	0.012	0.013
	Mol-1	0.051	0.200	0.200
	Mol-1C Sparse Epitope	**0.747**	**0.863**	**0.892**
	Mol-1C Oracle Epitope	0.678	0.857	0.874

C.2. Binding Kinetics

Multi-concentration kinetic analysis. BLI measurements used an Octet RH16 system (Sartorius) or a Gator Prime system (Gator Bio). For Octet assays, VHH-Fc was captured on ProA biosensors (Sartorius, catalogue no. 18-5012) in 0.05% PBST and exposed to serial antigen dilutions, typically approximately 16–1,000 nM. SPR measurements used a Biacore 8K system with a Protein A sensor chip (Cytiva, catalogue no. 29127556) in HBS-EP+ buffer. Captured VHH-Fc was exposed to twofold antigen dilutions spanning approximately 0.39–13 nM for BCMA or 1.6–50 nM for 4-1BB, with 120 s association and 200 s dissociation; surfaces were regenerated with glycine buffer (pH 1.5). Sensorgrams were globally fitted with a 1:1 binding model or, for cytokine-A/B measurements on the Gator system, a 2:1 heterogeneous-ligand model using approximately 16–500 nM antigen. Fits yielded association and dissociation rate constants and apparent K_D values in the VHH-Fc format. Steady-state affinity analysis was also used for selected interactions.

C.3. Ligand–Antibody Competition

Sequential BLI competition assays were performed in-house on a Gator Prime system (Gator Bio) with Anti-His (HIS) XT biosensors (Gator Bio, catalogue no. 160050) and 0.05% PBST. Biosensors were loaded with 10 μg mL^{-1} TNFR2-His (Sino Biological, catalogue no. 10417-H08H) or IL-4Rα-His (Sino Biological, catalogue no. 10402-H08H). TNFR2-loaded sensors were exposed to 250 nM TNF-α or assay buffer for 180 s and then to 200 nM TNFR2 VHH 01 for 180 s. IL-4Rα-loaded sensors were exposed to 100 nM IL-4 or buffer for 180 s and then to 100 nM IL-4Rα VHH 01 for 180 s. Antibody-binding responses after ligand exposure were compared with buffer-preexposed controls.

C.4. Cell-Surface Binding

GPCR cell-surface binding assays. Stable HEK293T cell lines expressing GFP-tagged CXCR6, CXCR4, or CCR5 were generated using the pLVX-Puro lentiviral vector. Lentiviral particles were produced by cotransfecting HEK293T packaging cells with the expression construct, psPAX2, and pMD2.G using polyethylenimine (PEI). Virus-containing supernatants were collected 48 h after transfection and used to infect HEK293T cells in the presence of 10 μg mL^{-1} polybrene. Stable cell lines were established by puromycin selection.

GPCR-expressing HEK293T cells and HEK293-null controls were tested in parallel. Cells were detached and resuspended in PBS containing 2% FBS and 2 mM EDTA, incubated with 250 nM VHH-Fc at 4 °C for 20 min, pelleted, and washed once. Bound VHH-Fc was detected with Alexa Fluor 647-conjugated goat anti-human IgG Fc secondary antibody (Thermo Fisher) at 4 °C for 20 min. After washing, cells were analyzed on a CytoFLEX flow cytometer (Beckman Coulter). Candidates were classified as positive when median Alexa Fluor 647 fluorescence intensity on GPCR-expressing cells exceeded three times that on HEK293-null cells under the same staining conditions.

For a selected positive VHH-Fc candidate, concentration-dependent binding was measured using eight twofold serial dilutions from 250 to 1.95 nM, with the same staining and flow-cytometry procedure. Median Alexa Fluor 647 fluorescence intensity of GFP-positive cells was fitted against VHH-Fc concentration using a four-parameter logistic model to determine the apparent cell-binding EC_{50}.

BCMA cell binding. Binding was measured by flow cytometry using human BCMA-expressing CHO-K1 cells (Genomeditech, catalogue no. GM-C47989), with 2×10^5 cells per well. Antibodies were tested at six fourfold serial dilutions from 12.5 to approximately 0.012 nM, with one well per concentration. Belantamab (Genomeditech, catalogue no. GM-52396AB) served as the positive reference and an unrelated VHH-Fc as the negative control. Bound antibody was detected with Alexa Fluor 647-conjugated goat anti-human IgG Fcγ-specific antibody (Jackson ImmunoResearch, catalogue no. 109-605-098; 1:800 dilution). Median APC fluorescence was fitted with a fixed-background Hill model to estimate EC_{50} over the same concentration range for all antibodies.

C.5. Functional Assays

GloSensor cAMP assay. Antagonist activity of VHH-Fc candidates was assessed by measuring reversal of ligand-induced inhibition of forskolin-stimulated cAMP accumulation. HEK293T cells stably expressing the target GPCR were transfected with the pGloSensor-22F cAMP plasmid (Promega). Before treatment, cells were equilibrated with GloSensor cAMP Reagent (Promega) according to the manufacturer's instructions. Cells were preincubated with serial dilutions of VHH-Fc at 37 °C for 10 min, followed by the corresponding ligand for an additional 10 min before forskolin addition. Luminescence was measured using a Spark multimode microplate reader (Tecan) and normalized to the forskolin-only control, defined as 100%.

4-1BB receptor activation. FcγRIIb-expressing CHO-K1 cells (Genomeditech, catalogue no. GM-C16925) were seeded at 1×10^4 cells per well. The following day, human 4-1BB reporter 293 cells (Genomeditech, catalogue no. GM-C04832; 1.5×10^4 cells per well) and diluted antibodies were added in DMEM containing 1% FBS and 1% penicillin–streptomycin. After 6 h, luciferase activity was measured using the GMOne-Step 2.0 Luciferase Reporter Gene Assay

Kit (Genomeditech, catalogue no. GM-040513). Utomilumab (Genomeditech, catalogue no. GM-26840AB) served as the positive reference; an unrelated antibody and antibody-free coculture served as negative and basal controls. Designed VHHs and the negative-control antibody were tested at nine threefold serial dilutions starting at 200 nM, and Utomilumab at ten fourfold serial dilutions starting at 200 nM. All conditions were run in duplicate. Responses were summarized as mean ± SD, and concentration–response curves were fitted with a four-parameter logistic model to estimate EC_{50}.

4-1BB screening after experimental feedback. The second-round screen used the same coculture conditions and luciferase assay. Each of 51 designs was tested at 3, 0.3, and 0.03 nM, with one well per design at each concentration. The two parent VHHs, the negative-control antibody, and Utomilumab were tested in duplicate. Antibody-free coculture and reporter-cell-only wells were included as controls. Responses at 0.03 nM were compared with the mean Utomilumab response at the same concentration.

C.6. Developability Assessment

Hydrophobicity, thermal stability, self-interaction, and nonspecific binding were assessed by HIC, DSF, AC-SINS, and BVP assays, respectively. Reference antibodies included infliximab (Biointron, Cat. No. B21631112), bevacizumab (Biointron, Cat. No. B7424), bococizumab (Biointron, Cat. No. B21505605), gantenerumab (Biointron, Cat. No. B523901), and trastuzumab (Biointron, Cat. No. B7432).

Screening criteria and complete-case analysis. We applied a literature-informed retrospective panel, building on clinical-antibody biophysical screening (Jain et al., 2017). The operational criteria were HIC retention time < 10.1 min, reported T_{m1} > 60 °C, AC-SINS wavelength shift $\Delta\lambda$ < 10 nm, and BVP Score < 19.09. The HIC and BVP benchmarks of 11.53 min and 5.30, respectively, were obtained from the approved-antibody reference panel of Jain et al. (Jain et al., 2023) and mapped to local assay scales using ordinary least-squares regression with an intercept. HIC calibration used 21 valid reference-antibody pairs and the mapped cutoff was 10.1 min. BVP calibrations using four reference antibodies yielded cutoff of 19.09. BVP Scores were normalized within each batch to gantenerumab = 100. The reported T_{m1} > 60 °C criterion was informed by the VHH-oriented screening (Chai Discovery Team, 2025b). The AC-SINS criterion followed the favorable < 10 nm range reported by Bailly et al. (Bailly et al., 2020).

Self-interaction by AC-SINS. AffiniPure goat anti-human IgG Fcγ-specific capture antibody (Jackson ImmunoResearch, catalogue no. 109-005-098) was buffer-exchanged into sodium acetate and conjugated to gold nanoparticles (Zhongkekeyou, catalogue no. ZKKYZW-1-20) at room temperature. Conjugates were blocked with poly(ethylene glycol) methyl ether thiol (Sigma, catalogue no. 729140), collected by centrifugation, and resuspended in PBS (pH 7.4). Samples at 0.05 mg mL^{-1} were mixed with conjugates and incubated at room temperature. Absorbance spectra were measured on an Epoch 2 microplate reader (Agilent), and $\Delta\lambda$ was calculated relative to conjugates alone. Infliximab was the positive self-interaction control; bevacizumab and goat IgG (Jackson ImmunoResearch, catalogue no. 005-000-003) were negative controls.

Thermal stability by DSF. Intrinsic fluorescence-based DSF used a SuprDSF instrument (Applied Photophysics). Samples at 1 mg mL^{-1} in PBS were analyzed in duplicate. Samples

were heated from 20 to 100 °C at 1 °C min^{-1}. Intrinsic tryptophan and tyrosine fluorescence was monitored at 330 and 350 nm. Melting temperatures were determined from first-derivative peaks of the barycentric mean fluorescence denaturation curves, and unfolding onset temperatures were recorded.

Hydrophobicity by HIC. HIC used an LC-2050C HPLC system (Shimadzu) and a MabPac HIC-Butyl column (Thermo Scientific). Samples were diluted in PBS and 30 μg protein was injected per analysis. The column was held at 25 °C with a flow rate of 1 mL min^{-1}. Mobile phase A contained 1.5 M ammonium sulfate and 50 mM potassium phosphate (pH 7.0); mobile phase B contained 50 mM potassium phosphate and 25% isopropanol (pH 7.0). Proteins were eluted with a 30 min linear gradient from 0 to 80% phase B, followed by a high-organic wash and re-equilibration.

Nonspecific binding by BVP ELISA. Baculovirus particles were coated onto assay plates overnight at 4 °C. Plates were washed once and blocked for 1 h at room temperature. After three washes, VHH-Fc samples were added at 100 nM in technical triplicate and incubated for 1 h at room temperature. Plates were washed six times, followed by incubation with HRP-conjugated goat anti-human IgG-Fc antibody (Sigma, catalogue no. A0170) for 1 h at room temperature. After six further washes, fluorescence was developed using Amplex Red (Beyotime, catalogue no. ST010-25mg) at room temperature, protected from light, and measured at 565 nm using a SpectraMax i3X multimode reader (Molecular Devices). Infliximab and bevacizumab served as negative controls, and bococizumab and gantenerumab as positive controls. BVP Scores were calculated as 100 × the mean sample fluorescence divided by the mean gantenerumab fluorescence within the same batch.

C.7. Hit rate

Table 3 | Target-level screening hit rates.

Target	Assay	Design candidates	Binding hits	Hit rate (%)
PDGFR	BLI	30	19	63
14-3-3ϵ	BLI	30	16	53
IL-20	BLI	20	10	50
CCR8	Nanodisc BLI	50	22	44
UBC9	BLI	30	13	43
BCMA	BLI	30	12	40
Cytokine-A	BLI	30	11	37
CXCR4	Cell binding	50	16	32
Cytokine-B	BLI	30	7	23
SAE1	BLI	30	7	23
PD-L1	BLI	20	4	20
SORCN	BLI	30	6	20
IL-4Rα	BLI	30	6	20
TNFR2	BLI	30	3	10
4-1BB	BLI	30	2	7
CXCR6	Cell binding	50	3	6
CCR5	Cell binding	50	3	6